# model_2500 · All 11 actual policy observations · env 40 · full 60 s

Isaac Lab · physics/PD 400 Hz · env 40
DR · 0.00–60.00 s




MuJoCo · physics/PD 400 Hz · env 40
DR · 0.00–60.00 s




MuJoCo · physics/PD 800 Hz · env 40
DR · 0.00–60.00 s



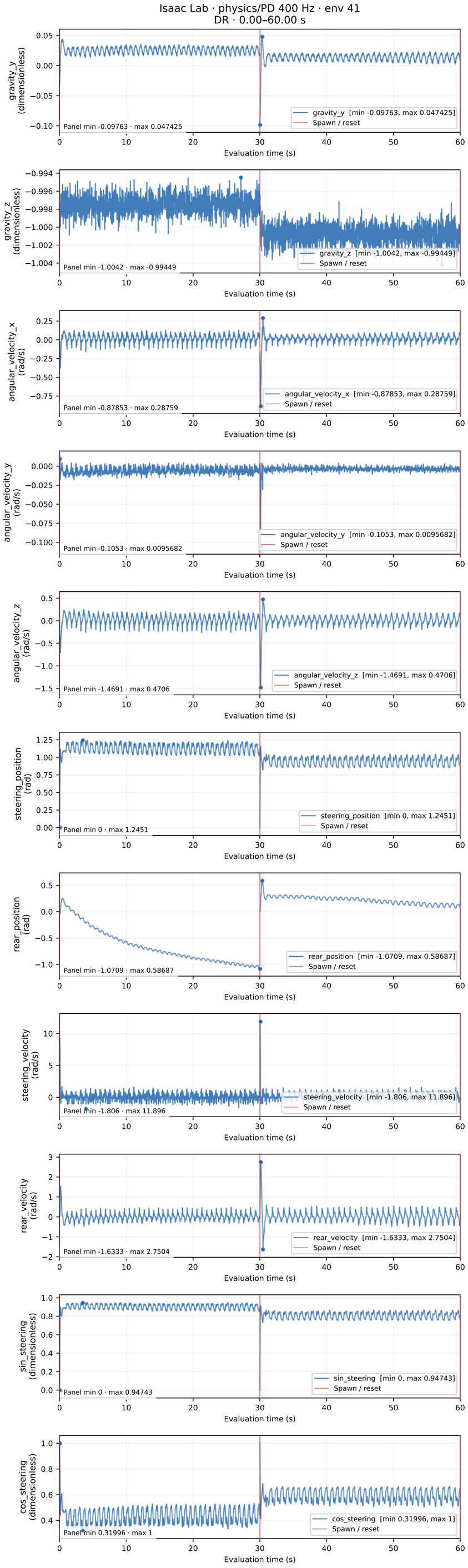

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

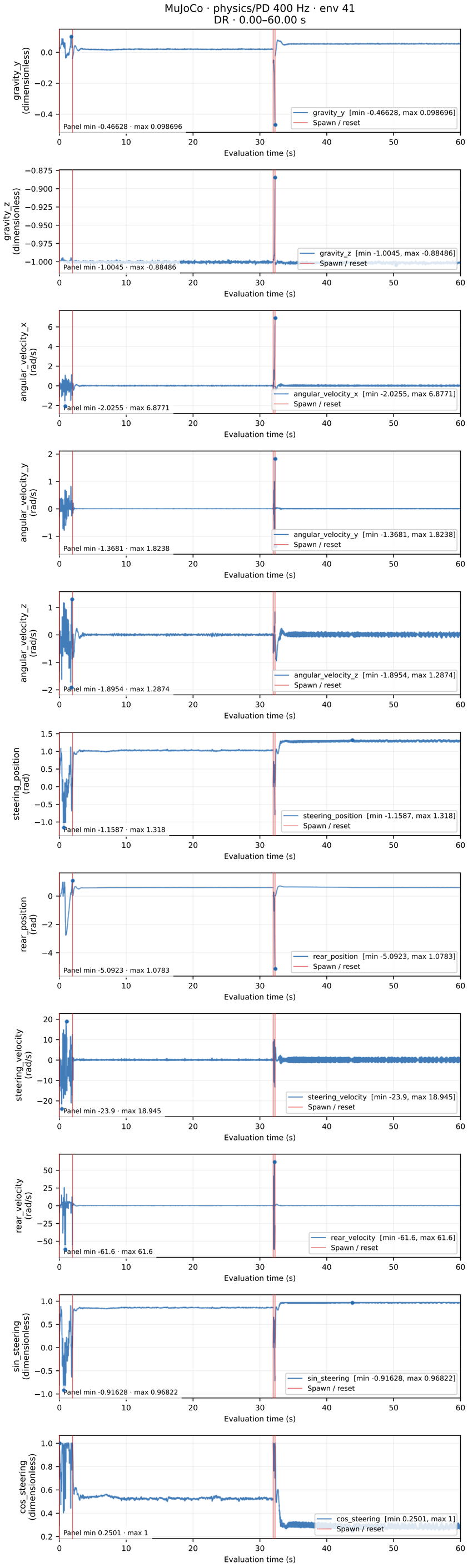
```
SAMPLED DR: env 40; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.04035, 1.04577, 0.979949, 1.08483]
  CoM XYZ mm [-0.0709146, -3.01796, 4.94215]
  mu_s [0.703711, 0.531594, 0.533455, 0.760821]
  mu_d [0.703711, 0.531594, 0.533455, 0.670315]
  Kp [steer,rear] [10.911, 0]; Kd [0.183236, 0]
  spawn roll -0.0712918 rad; rate 0.082492 rad/s
  gyro bias XYZ rad/s [-5.82015e-05, -0.000490921, -1.82122e-05]
  gravity bias YZ [0.000388784, 0.00204129]
E2 [30.00, 30.40) s:
  mass x [1.17788, 0.882942, 1.03983, 1.01883]
  CoM XYZ mm [2.16585, -4.89858, -4.63985]
  mu_s [0.764681, 0.774409, 0.760821, 0.632409]
  mu_d [0.764681, 0.561095, 0.670315, 0.589076]
  Kp [steer,rear] [10.5275, 0]; Kd [0.181964, 0]
  spawn roll 0.098748 rad; rate -0.121438 rad/s
  gyro bias XYZ rad/s [0.000241145, 0.000166553, -0.000184359]
  gravity bias YZ [0.00147458, -0.000643518]
E3 [30.40, 60.00) s:
  mass x [1.18893, 0.872782, 1.06795, 0.873044]
  CoM XYZ mm [4.9552, -0.0197962, 4.82924]
  mu_s [0.659445, 0.539956, 0.674963, 0.703711]
  mu_d [0.547619, 0.539956, 0.601294, 0.703711]
  Kp [steer,rear] [10.4112, 0]; Kd [0.200526, 0]
  spawn roll 0.069262 rad; rate -0.12872 rad/s
  gyro bias XYZ rad/s [-5.84255e-05, 0.000159831, -3.5917e-05]
  gravity bias YZ [0.000585921, -0.00253607]
Six significant digits shown; full precision: sampled_dr_values.json.
```

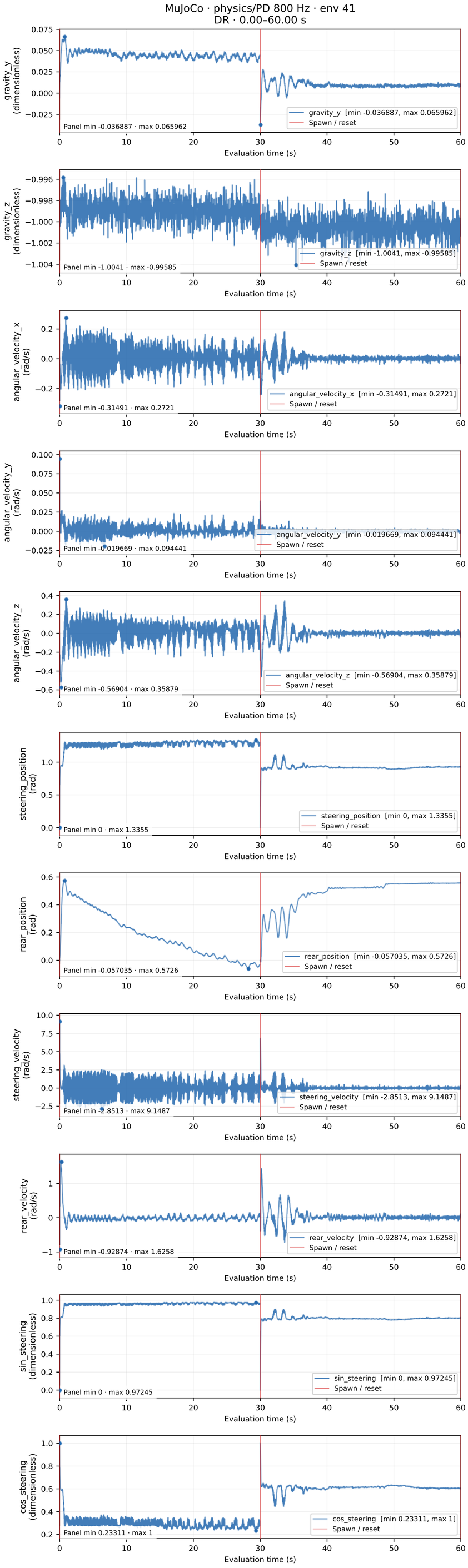
```
SAMPLED DR: env 40; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.19906, 1.14192, 0.887075, 1.19206]
  CoM XYZ mm [-0.91411, 4.39072, -2.76031]
  mu_s [0.757579, 0.748289, 0.666375, 0.700521]
  mu_d [0.70921, 0.689499, 0.519145, 0.641329]
  Kp [steer,rear] [9.28961, 0]; Kd [0.194382, 0]
  spawn roll 0.0207809 rad; rate -0.0459738 rad/s
  gyro bias XYZ rad/s [-0.000295331, 0.000226492, -0.000163432]
  gravity bias YZ [0.000325256, 0.000203647]
E2 [30.00, 60.00) s:
  mass x [0.888803, 1.02384, 0.93011, 1.12818]
  CoM XYZ mm [1.48571, 1.34925, 3.56072]
  mu_s [0.597748, 0.669571, 0.616244, 0.528253]
  mu_d [0.597748, 0.669571, 0.586498, 0.528253]
  Kp [steer,rear] [10.7976, 0]; Kd [0.180783, 0]
  spawn roll 0.0565158 rad; rate 0.209408 rad/s
  gyro bias XYZ rad/s [-5.45064e-05, -0.000171669, -0.000328325]
  gravity bias YZ [0.00244514, -0.000764611]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 40; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.19906, 1.14192, 0.887075, 1.19206]
  CoM XYZ mm [-0.91411, 4.39072, -2.76031]
  mu_s [0.757579, 0.748289, 0.666375, 0.700521]
  mu_d [0.70921, 0.689499, 0.519145, 0.641329]
  Kp [steer,rear] [9.28961, 0]; Kd [0.194382, 0]
  spawn roll 0.0207809 rad; rate -0.0459738 rad/s
  gyro bias XYZ rad/s [-0.000295331, 0.000226492, -0.000163432]
  gravity bias YZ [0.000325256, 0.000203647]
E2 [30.00, 60.00) s:
  mass x [0.888803, 1.02384, 0.93011, 1.12818]
  CoM XYZ mm [1.48571, 1.34925, 3.56072]
  mu_s [0.597748, 0.669571, 0.616244, 0.528253]
  mu_d [0.597748, 0.669571, 0.586498, 0.528253]
  Kp [steer,rear] [10.7976, 0]; Kd [0.180783, 0]
  spawn roll 0.0565158 rad; rate 0.209408 rad/s
  gyro bias XYZ rad/s [-5.45064e-05, -0.000171669, -0.000328325]
  gravity bias YZ [0.00244514, -0.000764611]
Six significant digits shown; full precision: sampled_dr_values.json.
```

# model_2500 · All 11 actual policy observations · env 41 · full 60 s



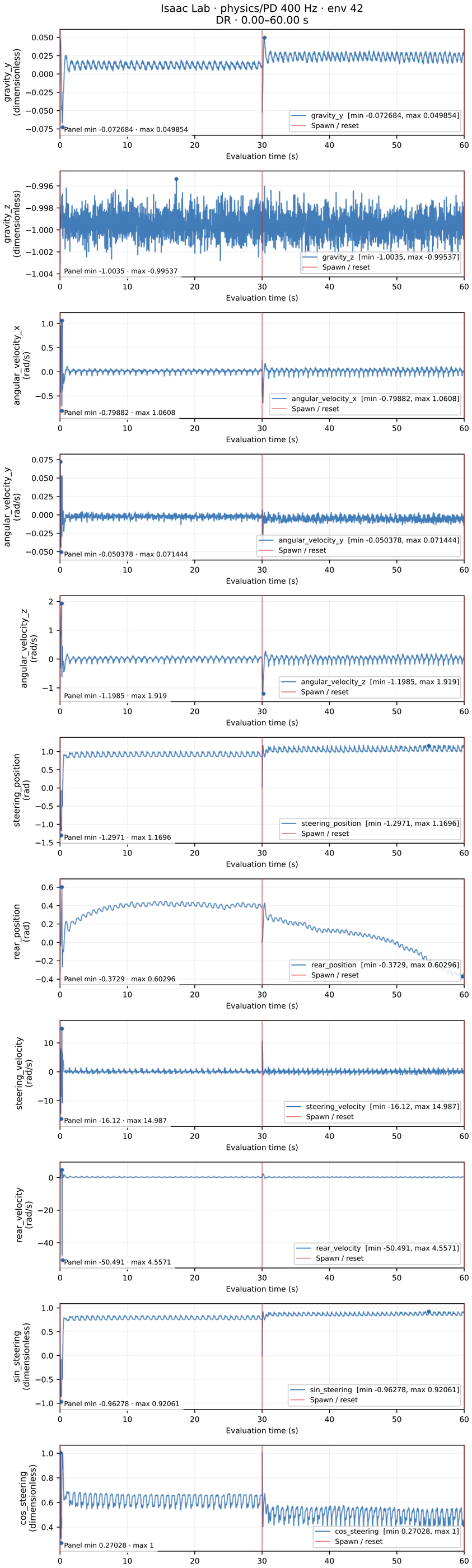

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

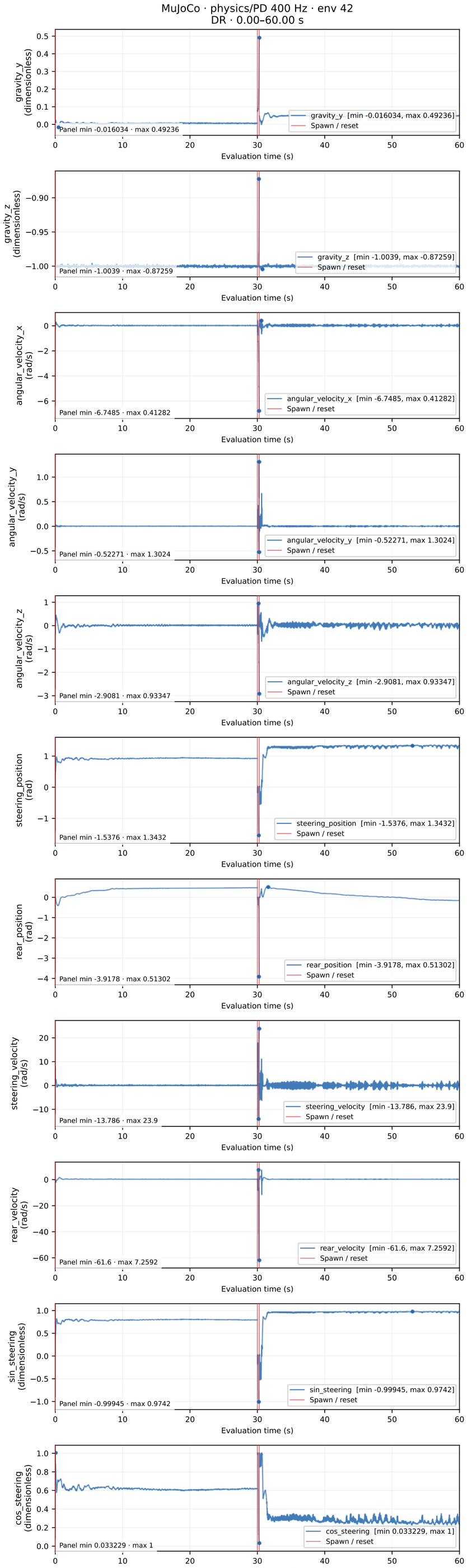
```
SAMPLED DR: env 41; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.02563, 0.983483, 1.08059, 0.972001]
  CoM XYZ mm [-3.28602, -0.157472, -3.32175]
  mu_s [0.764681, 0.693205, 0.559809, 0.674963]
  mu_d [0.764681, 0.693205, 0.559809, 0.601294]
  Kp [steer,rear] [10.9536, 0]; Kd [0.203215, 0]
  spawn roll 0.0304884 rad; rate -0.132846 rad/s
  gyro bias XYZ rad/s [4.11975e-05, -0.00018841, -8.61843e-05]
  gravity bias YZ [0.000214678, 0.00204717]
E2 [30.00, 60.00) s:
  mass x [0.975866, 1.08206, 0.843441, 1.14329]
  CoM XYZ mm [0.677288, 1.96078, -0.419378]
  mu_s [0.673378, 0.632409, 0.794338, 0.774409]
  mu_d [0.673378, 0.589076, 0.526209, 0.561095]
  Kp [steer,rear] [9.94711, 0]; Kd [0.207776, 0]
  spawn roll 0.0961761 rad; rate -0.169061 rad/s
  gyro bias XYZ rad/s [-3.30134e-06, -3.2802e-05, -0.000105036]
  gravity bias YZ [-0.000921459, -0.000930932]
Six significant digits shown; full precision: sampled_dr_values.json.
```

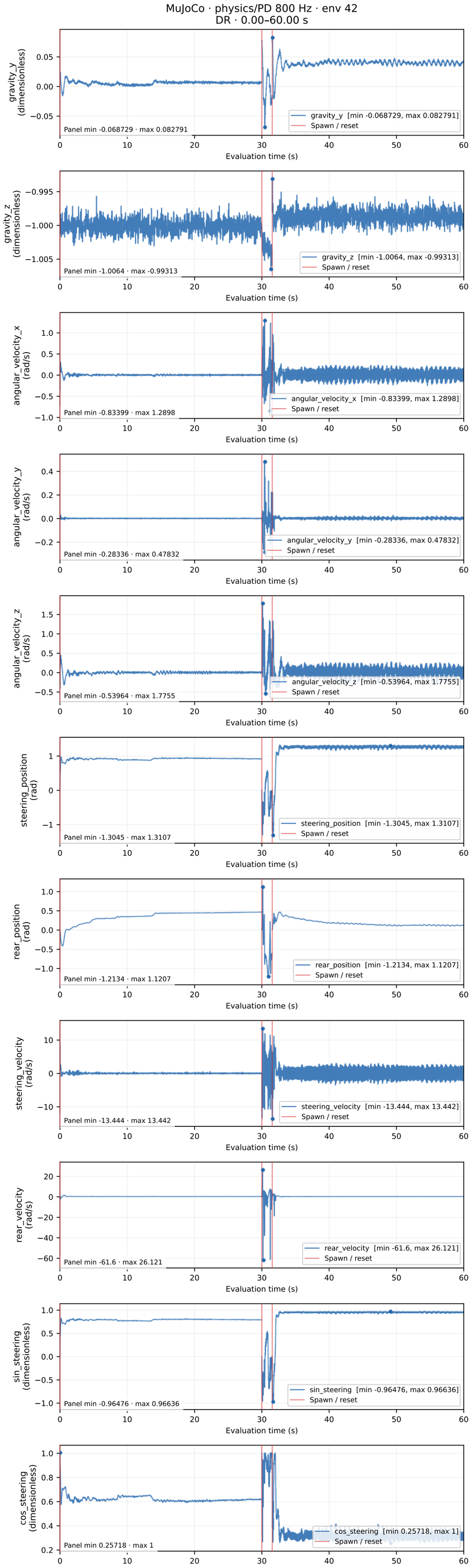
```
SAMPLED DR: env 41; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.98) s:
  mass x [0.926566, 0.817842, 1.05164, 0.922992]
  CoM XYZ mm [-1.5725, -2.52458, 3.87987]
  mu_s [0.640016, 0.640016, 0.666375, 0.791209]
  mu_d [0.513141, 0.513141, 0.519145, 0.767936]
  Kp [steer,rear] [10.1916, 0]; Kd [0.202588, 0]
  spawn roll 0.00910104 rad; rate 0.229638 rad/s
  gyro bias XYZ rad/s [-0.000103397, 0.00015139, -3.93015e-05]
  gravity bias YZ [0.00289244, 0.000194054]
E2 [1.98, 31.98) s:
  mass x [1.07076, 1.0927, 0.853399, 0.960507]
  CoM XYZ mm [-4.64046, 0.504941, 3.0578]
  mu_s [0.616244, 0.528253, 0.757579, 0.528253]
  mu_d [0.586498, 0.528253, 0.70921, 0.528253]
  Kp [steer,rear] [10.5888, 0]; Kd [0.197991, 0]
  spawn roll 0.0414456 rad; rate -0.120878 rad/s
  gyro bias XYZ rad/s [-0.000147001, -9.712e-05, -2.48645e-05]
  gravity bias YZ [-0.00115379, -0.000545909]
E3 [31.98, 32.26) s:
  mass x [0.876037, 0.815033, 0.989124, 1.15222]
  CoM XYZ mm [4.04001, 0.327695, 1.12065]
  mu_s [0.611239, 0.541939, 0.546287, 0.640867]
  mu_d [0.611239, 0.534359, 0.546287, 0.556841]
  Kp [steer,rear] [9.145, 0]; Kd [0.199156, 0]
  spawn roll 0.0656389 rad; rate -0.0234191 rad/s
  gyro bias XYZ rad/s [4.01662e-05, 0.000111145, -0.00016517]
  gravity bias YZ [-0.00235082, 0.00112893]
E4 [32.26, 60.00) s:
  mass x [1.16128, 1.118, 1.11372, 1.17773]
  CoM XYZ mm [-2.94626, -4.28865, -0.0523343]
  mu_s [0.733515, 0.528253, 0.568073, 0.749678]
  mu_d [0.558392, 0.528253, 0.568073, 0.741429]
  Kp [steer,rear] [9.90174, 0]; Kd [0.215294, 0]
  spawn roll 0.0237631 rad; rate -0.0780757 rad/s
  gyro bias XYZ rad/s [-7.63414e-05, 0.000195395, 0.000104616]
  gravity bias YZ [-0.000910715, -0.00288346]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 41; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [0.926566, 0.817842, 1.05164, 0.922992]
  CoM XYZ mm [-1.5725, -2.52458, 3.87987]
  mu_s [0.640016, 0.640016, 0.666375, 0.791209]
  mu_d [0.513141, 0.513141, 0.519145, 0.767936]
  Kp [steer,rear] [10.1916, 0]; Kd [0.202588, 0]
  spawn roll 0.00910104 rad; rate 0.229638 rad/s
  gyro bias XYZ rad/s [-0.000103397, 0.00015139, -3.93015e-05]
  gravity bias YZ [0.00289244, 0.000194054]
E2 [30.00, 60.00) s:
  mass x [1.05303, 0.809968, 1.18286, 0.98333]
  CoM XYZ mm [3.12944, 2.71152, 3.2709]
  mu_s [0.757579, 0.736077, 0.736077, 0.541939]
  mu_d [0.70921, 0.699455, 0.699455, 0.534359]
  Kp [steer,rear] [9.98297, 0]; Kd [0.216082, 0]
  spawn roll 0.039654 rad; rate -0.122531 rad/s
  gyro bias XYZ rad/s [-0.000155446, -0.000121868, 0.000286776]
  gravity bias YZ [0.000968121, -0.000573735]
Six significant digits shown; full precision: sampled_dr_values.json.
```

# model_2500 · All 11 actual policy observations · env 42 · full 60 s




50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

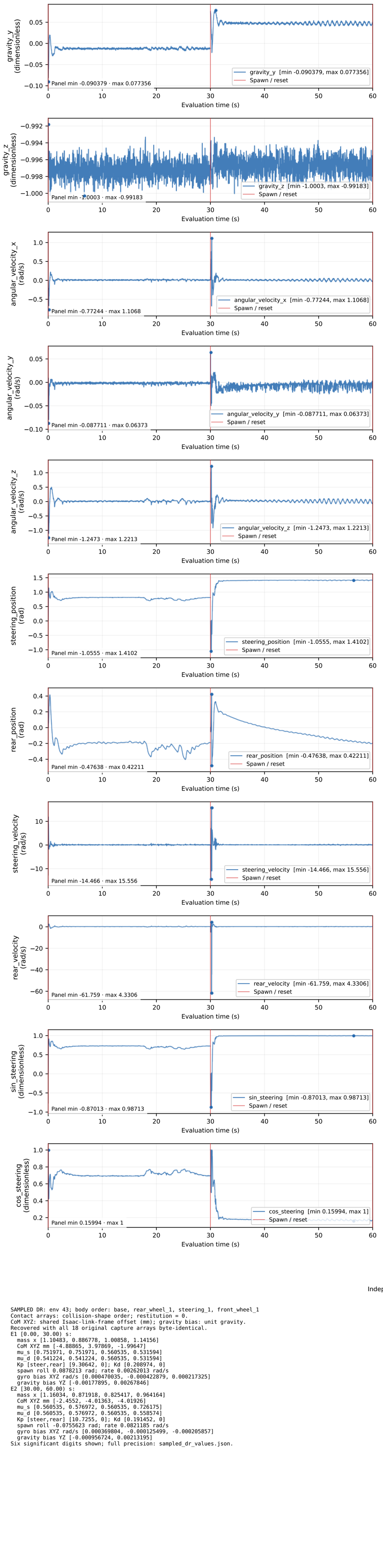
```
SAMPLED DR: env 42; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.00679, 1.17058, 1.15612, 1.14533]
  CoM XYZ mm [-3.16264, 2.93301, 3.42508]
  mu_s [0.560535, 0.713129, 0.703711, 0.598343]
  mu_d [0.560535, 0.713129, 0.703711, 0.598343]
  Kp [steer,rear] [9.3656, 0]; Kd [0.180648, 0]
  spawn roll -0.0355667 rad; rate -0.199113 rad/s
  gyro bias XYZ rad/s [7.18189e-05, 3.51286e-05, -4.70988e-05]
  gravity bias YZ [0.00497112, 0.000496189]
E2 [30.00, 60.00) s:
  mass x [1.11242, 1.10235, 0.950996, 0.941079]
  CoM XYZ mm [-1.10635, 0.569113, 2.59101]
  mu_s [0.785466, 0.764681, 0.673378, 0.693205]
  mu_d [0.52258, 0.764681, 0.673378, 0.693205]
  Kp [steer,rear] [10.383, 0]; Kd [0.183516, 0]
  spawn roll 0.0535442 rad; rate 0.212608 rad/s
  gyro bias XYZ rad/s [-0.000107776, -0.000431715, -2.87598e-05]
  gravity bias YZ [0.000301304, 0.000161024]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 42; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09284, 1.16008, 1.09972, 1.15357]
  CoM XYZ mm [-2.72544, 2.35069, 1.48796]
  mu_s [0.637675, 0.748289, 0.541939, 0.748289]
  mu_d [0.637675, 0.689499, 0.534359, 0.689499]
  Kp [steer,rear] [10.6795, 0]; Kd [0.195338, 0]
  spawn roll -0.0149055 rad; rate 0.134696 rad/s
  gyro bias XYZ rad/s [-0.000172796, -0.000281875, 0.000203764]
  gravity bias YZ [-0.00172276, -0.000162154]
E2 [30.00, 30.28) s:
  mass x [0.931213, 0.864553, 0.928057, 0.913452]
  CoM XYZ mm [-4.22538, 1.72408, -0.154362]
  mu_s [0.71008, 0.69316, 0.611239, 0.71155]
  mu_d [0.59371, 0.69316, 0.611239, 0.71155]
  Kp [steer,rear] [9.60562, 0]; Kd [0.212883, 0]
  spawn roll -0.0799131 rad; rate -0.0172209 rad/s
  gyro bias XYZ rad/s [0.000366462, 0.000184175, -0.000310861]
  gravity bias YZ [-0.00254312, -0.00370704]
E3 [30.28, 60.00) s:
  mass x [1.07123, 1.15427, 1.11965, 1.15157]
  CoM XYZ mm [-0.924762, -2.74748, 2.98176]
  mu_s [0.71008, 0.727426, 0.538976, 0.559972]
  mu_d [0.59371, 0.606358, 0.538976, 0.502209]
  Kp [steer,rear] [10.2255, 0]; Kd [0.213834, 0]
  spawn roll -0.0450423 rad; rate 0.0900405 rad/s
  gyro bias XYZ rad/s [-0.000205619, -2.14786e-05, -0.000159829]
  gravity bias YZ [0.0021959, -0.00092937]
Six significant digits shown; full precision: sampled_dr_values.json.
```

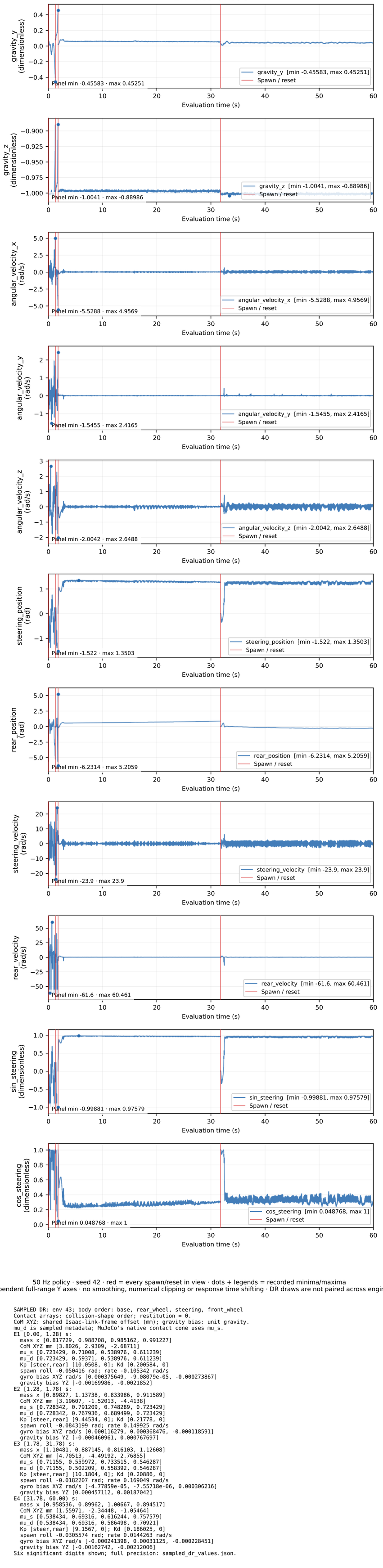
```
SAMPLED DR: env 42; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.09284, 1.16008, 1.09972, 1.15357]
  CoM XYZ mm [-2.72544, 2.35069, 1.48796]
  mu_s [0.637675, 0.748289, 0.541939, 0.748289]
  mu_d [0.637675, 0.689499, 0.534359, 0.689499]
  Kp [steer,rear] [10.6795, 0]; Kd [0.195338, 0]
  spawn roll -0.0149055 rad; rate 0.134696 rad/s
  gyro bias XYZ rad/s [-0.000172796, -0.000281875, 0.000203764]
  gravity bias YZ [-0.00172276, -0.000162154]
E2 [30.00, 31.54) s:
  mass x [0.931213, 0.864553, 0.928057, 0.913452]
  CoM XYZ mm [-4.22538, 1.72408, -0.154362]
  mu_s [0.71008, 0.69316, 0.611239, 0.71155]
  mu_d [0.59371, 0.69316, 0.611239, 0.71155]
  Kp [steer,rear] [9.60562, 0]; Kd [0.212883, 0]
  spawn roll -0.0799131 rad; rate -0.0172209 rad/s
  gyro bias XYZ rad/s [0.000366462, 0.000184175, -0.000310861]
  gravity bias YZ [-0.00254312, -0.00370704]
E3 [31.54, 60.00) s:
  mass x [1.17508, 0.936711, 0.962443, 0.975015]
  CoM XYZ mm [3.56882, -1.68577, -3.84009]
  mu_s [0.727426, 0.69316, 0.736077, 0.732187]
  mu_d [0.606358, 0.69316, 0.699455, 0.631664]
  Kp [steer,rear] [10.618, 0]; Kd [0.208953, 0]
  spawn roll -0.0754326 rad; rate -0.180929 rad/s
  gyro bias XYZ rad/s [0.000251584, -0.000109471, 0.0004679555]
  gravity bias YZ [0.000706018, 0.000487003]
Six significant digits shown; full precision: sampled_dr_values.json.
```

# model_2500 · All 11 actual policy observations · env 43 · full 60 s

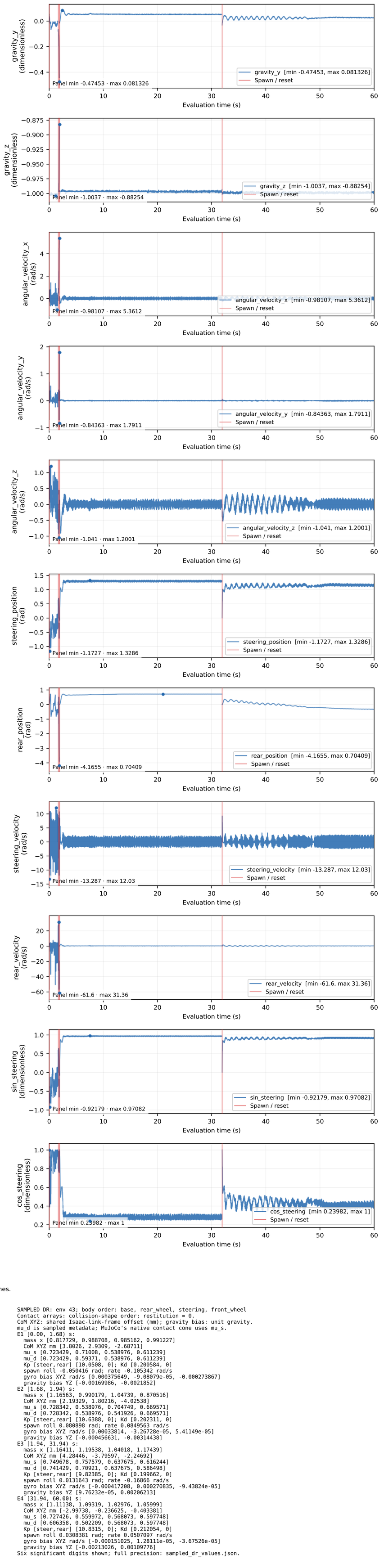
Isaac Lab · physics/PD 400 Hz · env 43
DR · 0.00–60.00 s




MuJoCo · physics/PD 400 Hz · env 43
DR · 0.00–60.00 s




MuJoCo · physics/PD 800 Hz · env 43
DR · 0.00–60.00 s




50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 43; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.10483, 0.886778, 1.00858, 1.14156]
  CoM XYZ mm [-4.88865, 3.97869, -1.99647]
  mu_s [0.751971, 0.751971, 0.560535, 0.531594]
  mu_d [0.541224, 0.541224, 0.560535, 0.531594]
  Kp [steer,rear] [9.30642, 0]; Kd [0.208974, 0]
  spawn roll 0.0878213 rad; rate 0.00262013 rad/s
  gyro bias XYZ rad/s [0.000470035, -0.000422879, 0.000217325]
  gravity bias YZ [-0.00177895, 0.00267846]
E2 [30.00, 60.00) s:
  mass x [1.16034, 0.871918, 0.825417, 0.964164]
  CoM XYZ mm [-2.4552, -4.01363, -4.01926]
  mu_s [0.560535, 0.576972, 0.560535, 0.726175]
  mu_d [0.560535, 0.576972, 0.560535, 0.558574]
  Kp [steer,rear] [10.7255, 0]; Kd [0.191452, 0]
  spawn roll -0.0755623 rad; rate 0.0821185 rad/s
  gyro bias XYZ rad/s [0.000369804, -0.000125499, -0.000205857]
  gravity bias YZ [-0.000956724, 0.00213195]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 43; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.28) s:
  mass x [0.817729, 0.988708, 0.985162, 0.991227]
  CoM XYZ mm [3.8026, 2.9309, -2.68711]
  mu_s [0.723429, 0.71008, 0.538976, 0.611239]
  mu_d [0.723429, 0.59371, 0.538976, 0.611239]
  Kp [steer,rear] [10.0508, 0]; Kd [0.200584, 0]
  spawn roll -0.050416 rad; rate -0.105342 rad/s
  gyro bias XYZ rad/s [0.000375649, -9.08079e-05, -0.000273867]
  gravity bias YZ [-0.00169986, -0.0021852]
E2 [1.28, 1.78) s:
  mass x [0.89827, 1.13738, 0.833986, 0.911589]
  CoM XYZ mm [3.19607, -1.52013, -4.4138]
  mu_s [0.728342, 0.791209, 0.748289, 0.723429]
  mu_d [0.728342, 0.767936, 0.689499, 0.723429]
  Kp [steer,rear] [9.44534, 0]; Kd [0.21778, 0]
  spawn roll -0.0843199 rad; rate 0.149925 rad/s
  gyro bias XYZ rad/s [0.000116279, 0.000368476, -0.000118591]
  gravity bias YZ [-0.000460961, 0.000767697]
E3 [1.78, 31.78) s:
  mass x [1.10481, 0.887145, 0.816103, 1.12608]
  CoM XYZ mm [4.70513, -4.49192, 2.76855]
  mu_s [0.71155, 0.559972, 0.733515, 0.546287]
  mu_d [0.71155, 0.502209, 0.558392, 0.546287]
  Kp [steer,rear] [10.1804, 0]; Kd [0.20886, 0]
  spawn roll -0.0182207 rad; rate 0.169049 rad/s
  gyro bias XYZ rad/s [-4.77859e-05, -7.55718e-06, 0.000306216]
  gravity bias YZ [0.000457112, 0.00187042]
E4 [31.78, 60.00) s:
  mass x [0.958536, 0.89962, 1.00667, 0.894517]
  CoM XYZ mm [1.55971, -2.34448, -1.05464]
  mu_s [0.538434, 0.69316, 0.616244, 0.757579]
  mu_d [0.538434, 0.69316, 0.586498, 0.70921]
  Kp [steer,rear] [9.1567, 0]; Kd [0.186025, 0]
  spawn roll -0.0305574 rad; rate 0.0144263 rad/s
  gyro bias XYZ rad/s [-0.000241398, 0.00031125, -0.000228451]
  gravity bias YZ [-0.00162742, -0.00212006]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 43; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.68) s:
  mass x [0.817729, 0.988708, 0.985162, 0.991227]
  CoM XYZ mm [3.8026, 2.9309, -2.68711]
  mu_s [0.723429, 0.71008, 0.538976, 0.611239]
  mu_d [0.723429, 0.59371, 0.538976, 0.611239]
  Kp [steer,rear] [10.0508, 0]; Kd [0.200584, 0]
  spawn roll -0.050416 rad; rate -0.105342 rad/s
  gyro bias XYZ rad/s [0.000375649, -9.08079e-05, -0.000273867]
  gravity bias YZ [-0.00169986, -0.0021852]
E2 [1.68, 1.94) s:
  mass x [1.16563, 0.990179, 1.04739, 0.870516]
  CoM XYZ mm [2.19329, 1.80216, -4.02538]
  mu_s [0.728342, 0.538976, 0.704749, 0.669571]
  mu_d [0.728342, 0.538976, 0.541926, 0.669571]
  Kp [steer,rear] [10.6388, 0]; Kd [0.202311, 0]
  spawn roll 0.080898 rad; rate 0.0849563 rad/s
  gyro bias XYZ rad/s [0.00033814, -3.26728e-05, 5.41149e-05]
  gravity bias YZ [-0.000456631, -0.00314438]
E3 [1.94, 31.94) s:
  mass x [1.16411, 1.19538, 1.04018, 1.17439]
  CoM XYZ mm [4.28446, -3.79597, -2.24692]
  mu_s [0.749678, 0.757579, 0.637675, 0.616244]
  mu_d [0.741429, 0.70921, 0.637675, 0.586498]
  Kp [steer,rear] [9.82385, 0]; Kd [0.199662, 0]
  spawn roll 0.0131643 rad; rate -0.16866 rad/s
  gyro bias XYZ rad/s [-0.000417208, 0.000270835, -9.43824e-05]
  gravity bias YZ [9.76232e-05, 0.00206213]
E4 [31.94, 60.00) s:
  mass x [1.11138, 1.09319, 1.02976, 1.05999]
  CoM XYZ mm [-2.99738, -0.236625, -0.403381]
  mu_s [0.727426, 0.559972, 0.568073, 0.597748]
  mu_d [0.606358, 0.502209, 0.568073, 0.597748]
  Kp [steer,rear] [10.8315, 0]; Kd [0.212054, 0]
  spawn roll 0.0308381 rad; rate 0.0507097 rad/s
  gyro bias XYZ rad/s [-0.000151025, 1.28111e-05, -3.67526e-05]
  gravity bias YZ [-0.00213026, 0.00109776]
Six significant digits shown; full precision: sampled_dr_values.json.
```

# model_2500 · All 11 actual policy observations · env 44 · full 60 s









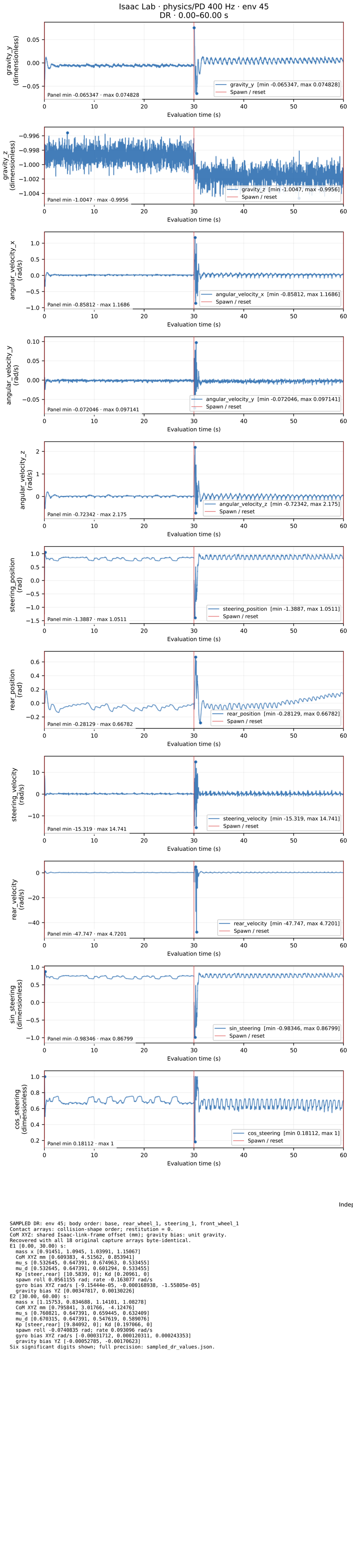

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 44; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 13.94) s:
  mass x [0.941381, 0.96656, 0.849268, 0.883645]
  CoM XYZ mm [-3.95469, -0.882261, -4.4395]
  mu_s [0.647391, 0.726175, 0.533455, 0.726175]
  mu_d [0.647391, 0.558574, 0.533455, 0.558574]
  Kp [steer,rear] [10.553, 0]; Kd [0.203251, 0]
  spawn roll -0.0903528 rad; rate -0.239559 rad/s
  gyro bias XYZ rad/s [-0.000238181, -9.13169e-05, -0.000286138]
  gravity bias YZ [0.00360941, -0.00105445]
E2 [13.94, 43.94) s:
  mass x [1.14842, 0.849827, 1.19386, 1.02724]
  CoM XYZ mm [-4.23548, -3.53206, -0.988275]
  mu_s [0.533455, 0.787792, 0.547743, 0.774409]
  mu_d [0.533455, 0.617134, 0.547743, 0.561095]
  Kp [steer,rear] [10.1177, 0]; Kd [0.192887, 0]
  spawn roll 0.0182498 rad; rate -0.0147221 rad/s
  gyro bias XYZ rad/s [-0.000116376, 0.000249171, 0.000123057]
  gravity bias YZ [1.07443e-05, 0.00194308]
E3 [43.94, 48.48) s:
  mass x [1.15761, 1.16175, 1.02536, 1.17213]
  CoM XYZ mm [-2.67453, -4.62616, 4.68497]
  mu_s [0.598343, 0.647391, 0.533455, 0.647391]
  mu_d [0.598343, 0.647391, 0.533455, 0.647391]
  Kp [steer,rear] [10.7051, 0]; Kd [0.203398, 0]
  spawn roll -0.0753268 rad; rate 0.106684 rad/s
  gyro bias XYZ rad/s [-0.000162301, -0.000414313, 7.95857e-05]
  gravity bias YZ [0.00263486, -0.00279752]
E4 [48.48, 60.00) s:
  mass x [1.18532, 1.03488, 1.18563, 1.19369]
  CoM XYZ mm [-2.79817, 1.75211, -3.49937]
  mu_s [0.774409, 0.579974, 0.726175, 0.674963]
  mu_d [0.561095, 0.579974, 0.558574, 0.601294]
  Kp [steer,rear] [9.84255, 0]; Kd [0.180638, 0]
  spawn roll -0.00713564 rad; rate 0.0294966 rad/s
  gyro bias XYZ rad/s [5.65455e-05, -0.000190306, -0.000184131]
  gravity bias YZ [5.0621e-05, -0.000538262]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 44; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.46) s:
  mass x [1.04618, 0.870142, 1.04623, 0.949018]
  CoM XYZ mm [-1.31651, 2.93645, 4.59471]
  mu_s [0.538434, 0.538434, 0.736077, 0.733515]
  mu_d [0.538434, 0.538434, 0.699455, 0.558392]
  Kp [steer,rear] [9.68445, 0]; Kd [0.190976, 0]
  spawn roll -0.0585563 rad; rate 0.157051 rad/s
  gyro bias XYZ rad/s [1.85104e-05, -0.000490205, 0.000237956]
  gravity bias YZ [0.00337487, -7.0869e-05]
E2 [0.46, 30.46) s:
  mass x [0.954605, 0.835809, 1.07741, 1.06988]
  CoM XYZ mm [-4.10035, 0.884504, 4.14273]
  mu_s [0.728342, 0.736077, 0.69316, 0.546287]
  mu_d [0.728342, 0.699455, 0.69316, 0.546287]
  Kp [steer,rear] [9.26279, 0]; Kd [0.187364, 0]
  spawn roll 0.0845821 rad; rate -0.0686739 rad/s
  gyro bias XYZ rad/s [-0.000148312, -4.68555e-05, 0.000189854]
  gravity bias YZ [-0.000816323, -0.00182729]
E3 [30.46, 60.00) s:
  mass x [0.990045, 1.03445, 0.878945, 1.18026]
  CoM XYZ mm [-1.52486, -3.60773, 3.41697]
  mu_s [0.631146, 0.757579, 0.749678, 0.749678]
  mu_d [0.631146, 0.70921, 0.741429, 0.741429]
  Kp [steer,rear] [9.51207, 0]; Kd [0.201351, 0]
  spawn roll -0.00503864 rad; rate 0.0751251 rad/s
  gyro bias XYZ rad/s [5.75694e-06, -0.000395627, -0.000485063]
  gravity bias YZ [0.00123184, -0.005065]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 44; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.04618, 0.870142, 1.04623, 0.949018]
  CoM XYZ mm [-1.31651, 2.93645, 4.59471]
  mu_s [0.538434, 0.538434, 0.736077, 0.733515]
  mu_d [0.538434, 0.538434, 0.699455, 0.558392]
  Kp [steer,rear] [9.68445, 0]; Kd [0.190976, 0]
  spawn roll -0.0585563 rad; rate 0.157051 rad/s
  gyro bias XYZ rad/s [1.85104e-05, -0.000490205, 0.000237956]
  gravity bias YZ [0.00337487, -7.0869e-05]
E2 [30.00, 30.38) s:
  mass x [0.948621, 1.00576, 1.15524, 1.08679]
  CoM XYZ mm [3.67924, 1.66306, 1.97143]
  mu_s [0.791209, 0.528253, 0.704749, 0.631146]
  mu_d [0.767936, 0.528253, 0.541926, 0.631146]
  Kp [steer,rear] [9.56033, 0]; Kd [0.181751, 0]
  spawn roll -0.0792774 rad; rate 0.0337022 rad/s
  gyro bias XYZ rad/s [3.46728e-05, -5.81238e-05, 0.000143752]
  gravity bias YZ [0.0014215, -5.02601e-05]
E3 [30.38, 60.00) s:
  mass x [0.909688, 1.1506, 0.868769, 0.976718]
  CoM XYZ mm [3.58117, -2.42407, -0.289591]
  mu_s [0.732187, 0.637675, 0.757579, 0.732187]
  mu_d [0.631664, 0.637675, 0.70921, 0.631664]
  Kp [steer,rear] [10.997, 0]; Kd [0.189741, 0]
  spawn roll 0.00442566 rad; rate 0.0122795 rad/s
  gyro bias XYZ rad/s [-0.000221453, -0.000250809, 5.67582e-05]
  gravity bias YZ [-0.00189388, -0.00247438]
Six significant digits shown; full precision: sampled_dr_values.json.
```

# model_2500 · All 11 actual policy observations · env 45 · full 60 s



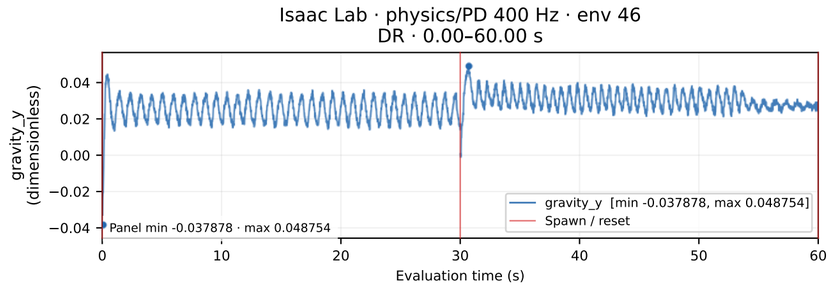

50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

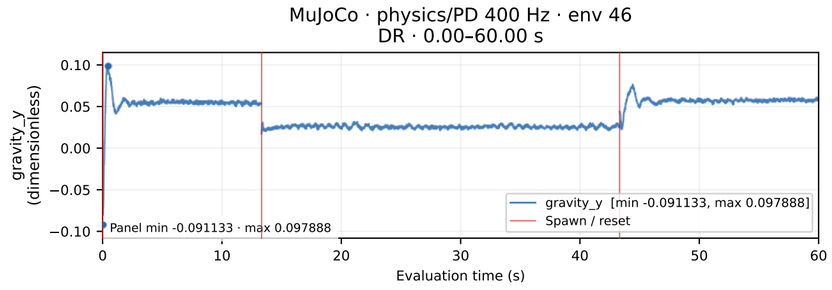
```
SAMPLED DR: env 45; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.91451, 1.0945, 1.03991, 1.15067]
  CoM XYZ mm [0.609383, 4.51562, 0.853941]
  mu_s [0.532645, 0.647391, 0.674963, 0.533455]
  mu_d [0.532645, 0.647391, 0.601294, 0.533455]
  Kp [steer,rear] [10.5839, 0]; Kd [0.20961, 0]
  spawn roll 0.0561155 rad; rate -0.163077 rad/s
  gyro bias XYZ rad/s [-9.15444e-05, -0.000168938, -1.55805e-05]
  gravity bias YZ [0.00347817, 0.00130226]
E2 [30.00, 60.00) s:
  mass x [1.15753, 0.834688, 1.14101, 1.08278]
  CoM XYZ mm [0.795841, 3.01766, -4.12476]
  mu_s [0.760821, 0.647391, 0.659445, 0.632409]
  mu_d [0.670315, 0.647391, 0.547619, 0.589076]
  Kp [steer,rear] [9.84092, 0]; Kd [0.197066, 0]
  spawn roll -0.0740835 rad; rate 0.093096 rad/s
  gyro bias XYZ rad/s [-0.00031712, 0.000120311, 0.000243353]
  gravity bias YZ [-0.00052785, -0.00170623]
Six significant digits shown; full precision: sampled_dr_values.json.
```

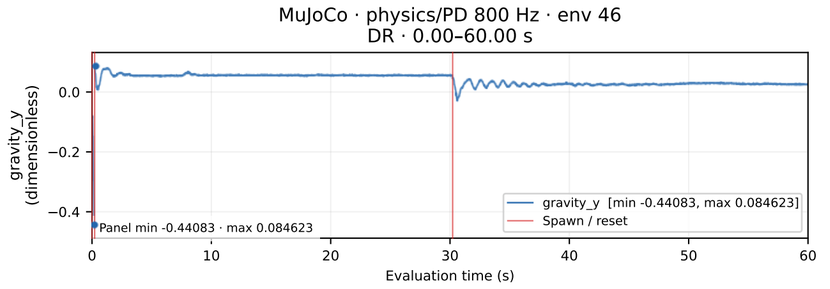
```
SAMPLED DR: env 45; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.60) s:
  mass x [1.0283, 1.11342, 0.868957, 0.992338]
  CoM XYZ mm [-0.719669, -3.11429, 0.391038]
  mu_s [0.616244, 0.597748, 0.736077, 0.616244]
  mu_d [0.586498, 0.597748, 0.699455, 0.586498]
  Kp [steer,rear] [9.60248, 0]; Kd [0.180136, 0]
  spawn roll -0.090286 rad; rate -0.0511712 rad/s
  gyro bias XYZ rad/s [0.000372074, 1.53988e-05, -0.000226969]
  gravity bias YZ [0.0011662, -0.00273945]
E2 [0.60, 0.86) s:
  mass x [0.84832, 1.17614, 0.915474, 0.899313]
  CoM XYZ mm [1.10572, -3.69804, 0.460751]
  mu_s [0.559972, 0.732187, 0.559972, 0.71155]
  mu_d [0.502209, 0.631664, 0.502209, 0.71155]
  Kp [steer,rear] [9.34733, 0]; Kd [0.19494, 0]
  spawn roll 0.0842819 rad; rate 0.176581 rad/s
  gyro bias XYZ rad/s [0.000186427, -9.71195e-05, -4.72349e-05]
  gravity bias YZ [0.00366361, 0.0010572]
E3 [0.86, 1.46) s:
  mass x [1.10776, 0.995629, 1.1679, 1.04241]
  CoM XYZ mm [-3.56775, 1.71396, -2.33411]
  mu_s [0.69316, 0.69316, 0.700521, 0.546287]
  mu_d [0.69316, 0.69316, 0.641329, 0.546287]
  Kp [steer,rear] [9.55618, 0]; Kd [0.185214, 0]
  spawn roll -0.0837655 rad; rate -0.190753 rad/s
  gyro bias XYZ rad/s [9.42402e-05, -0.000129821, -0.000217799]
  gravity bias YZ [-0.00055415, -0.00216323]
E4 [1.46, 2.10) s:
  mass x [0.936873, 0.87709, 0.844336, 0.859617]
  CoM XYZ mm [-0.365105, 2.8358, 4.34363]
  mu_s [0.541939, 0.757579, 0.791209, 0.757579]
  mu_d [0.534359, 0.70921, 0.767936, 0.70921]
  Kp [steer,rear] [9.17322, 0]; Kd [0.218362, 0]
  spawn roll -0.00954031 rad; rate 0.0414334 rad/s
  gyro bias XYZ rad/s [-0.000447509, -0.000180606, 8.89018e-05]
  gravity bias YZ [0.00342217, -6.01172e-05]
E5 [2.10, 32.10) s:
  mass x [0.928924, 1.12088, 1.17138, 0.908778]
  CoM XYZ mm [-0.303269, -4.07428, -0.365249]
  mu_s [0.640867, 0.732187, 0.541939, 0.791209]
  mu_d [0.556841, 0.631664, 0.534359, 0.767936]
  Kp [steer,rear] [10.7022, 0]; Kd [0.180027, 0]
  spawn roll 0.00338899 rad; rate 0.21835 rad/s
  gyro bias XYZ rad/s [5.8181e-05, 0.000126456, 9.24898e-05]
  gravity bias YZ [-0.00237553, -0.000891357]
E6 [32.10, 60.00) s:
  mass x [1.07428, 1.03262, 1.13889, 0.939128]
  CoM XYZ mm [-3.6004, 1.69268, -3.89672]
  mu_s [0.71008, 0.538434, 0.640867, 0.611239]
  mu_d [0.59371, 0.538434, 0.556841, 0.611239]
  Kp [steer,rear] [10.6163, 0]; Kd [0.192153, 0]
  spawn roll 0.085458 rad; rate -0.190884 rad/s
  gyro bias XYZ rad/s [-0.000152866, -0.000116929, -0.000136895]
  gravity bias YZ [-0.000396986, -0.0041253]
Six significant digits shown; full precision: sampled_dr_values.json.
```

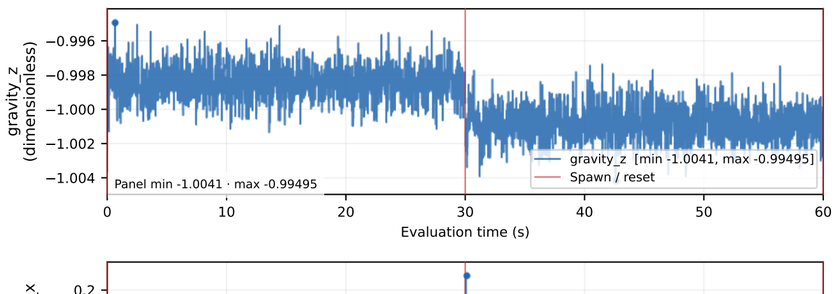
```
SAMPLED DR: env 45; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 30.00) s:
  mass x [1.0283, 1.11342, 0.868957, 0.992338]
  CoM XYZ mm [-0.719669, -3.11429, 0.391038]
  mu_s [0.616244, 0.597748, 0.736077, 0.616244]
  mu_d [0.586498, 0.597748, 0.699455, 0.586498]
  Kp [steer,rear] [9.60248, 0]; Kd [0.180136, 0]
  spawn roll -0.090286 rad; rate -0.0511712 rad/s
  gyro bias XYZ rad/s [0.000372074, 1.53988e-05, -0.000226969]
  gravity bias YZ [0.0011662, -0.00273945]
E2 [30.00, 60.00) s:
  mass x [0.979572, 1.07169, 1.03473, 1.18028]
  CoM XYZ mm [0.736894, -4.09673, 1.10124]
  mu_s [0.748289, 0.748289, 0.611239, 0.637675]
  mu_d [0.689499, 0.689499, 0.611239, 0.637675]
  Kp [steer,rear] [9.91425, 0]; Kd [0.191506, 0]
  spawn roll 0.0225475 rad; rate -0.237895 rad/s
  gyro bias XYZ rad/s [0.000177359, -0.000212698, 7.98098e-05]
  gravity bias YZ [-0.00389648, -0.000647599]
Six significant digits shown; full precision: sampled_dr_values.json.
```

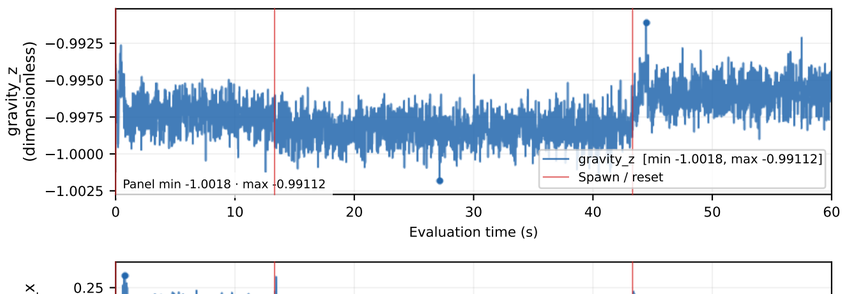
# model_2500 · All 11 actual policy observations · env 46 · full 60 s

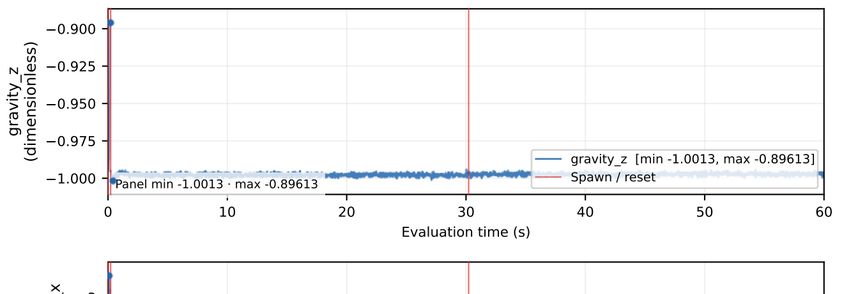


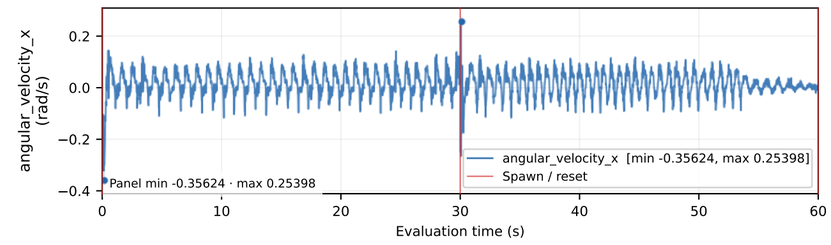



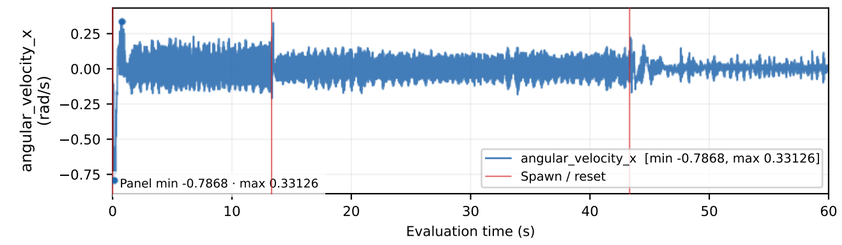



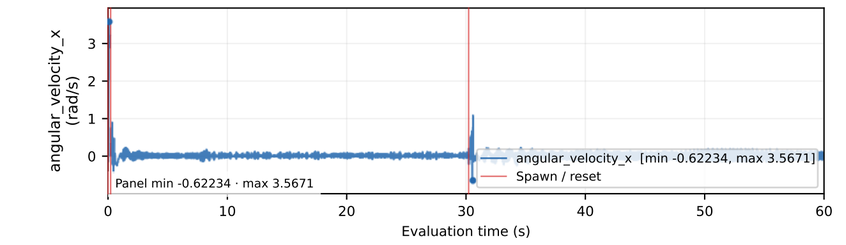



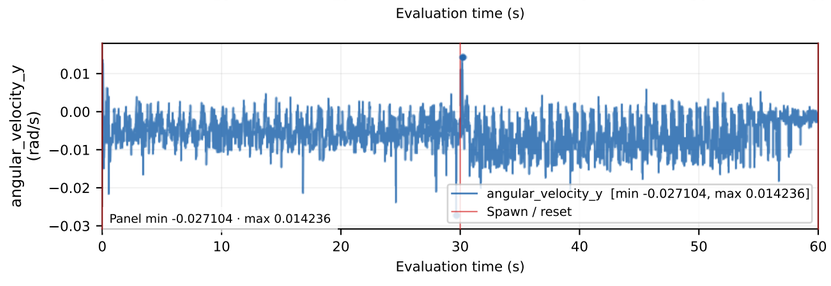



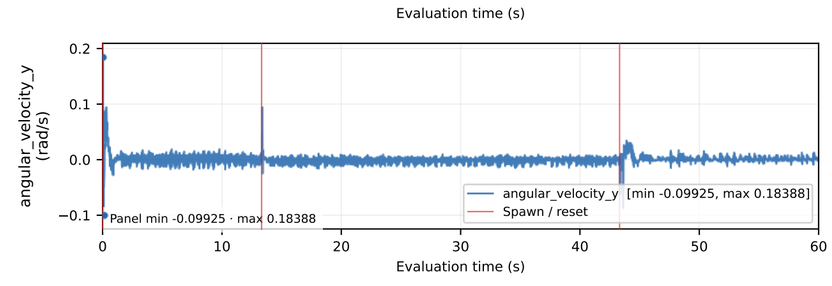



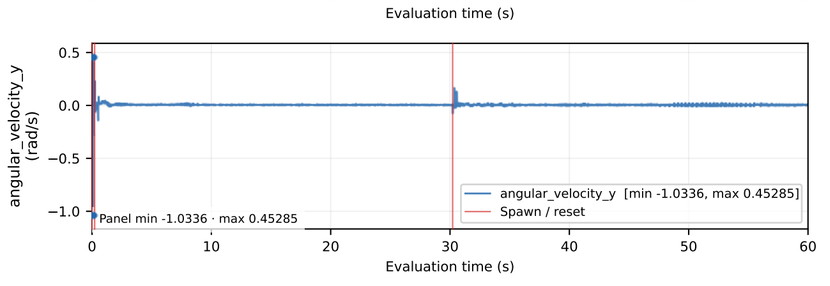



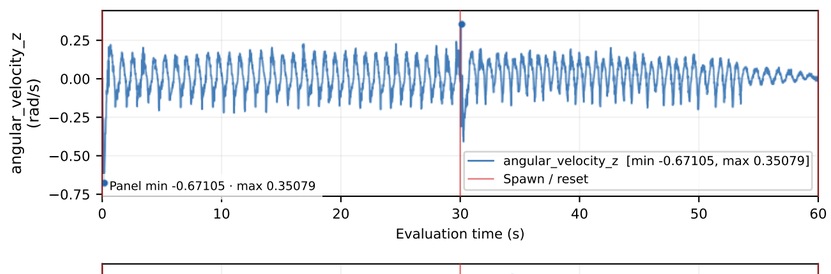



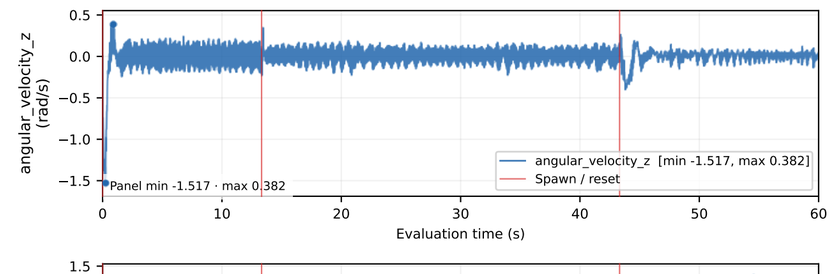



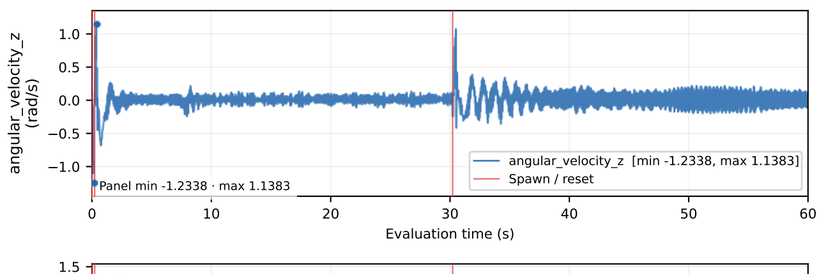



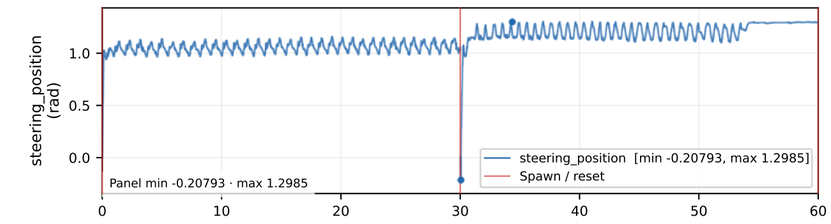



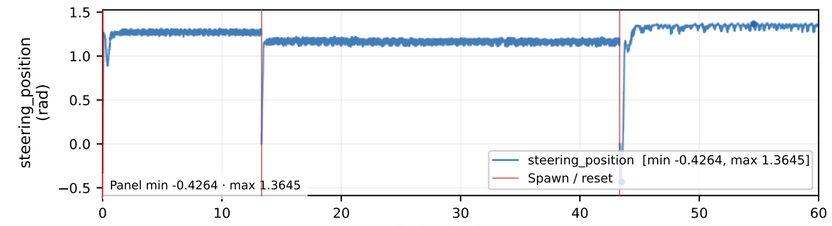



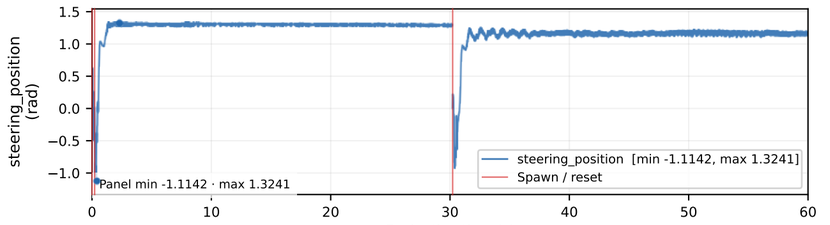



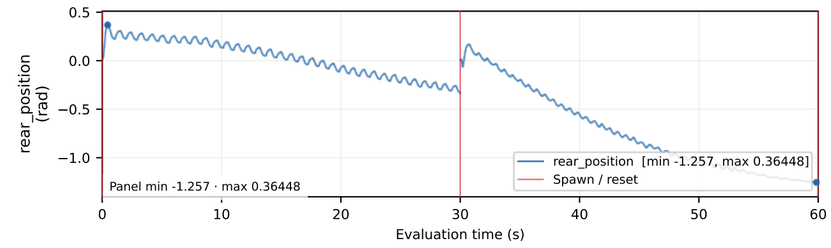



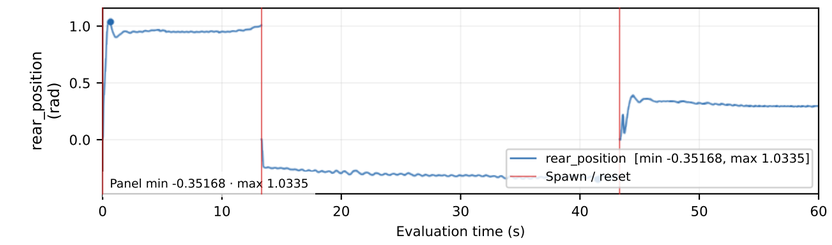



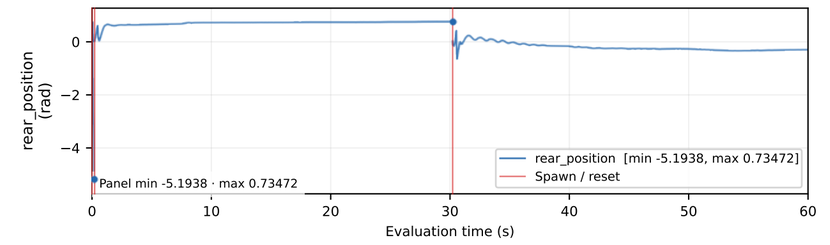



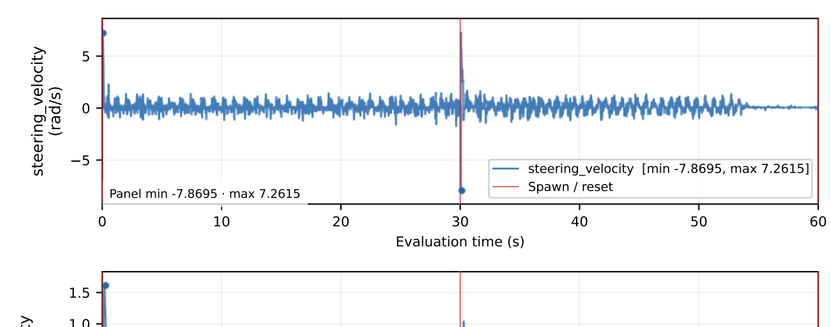



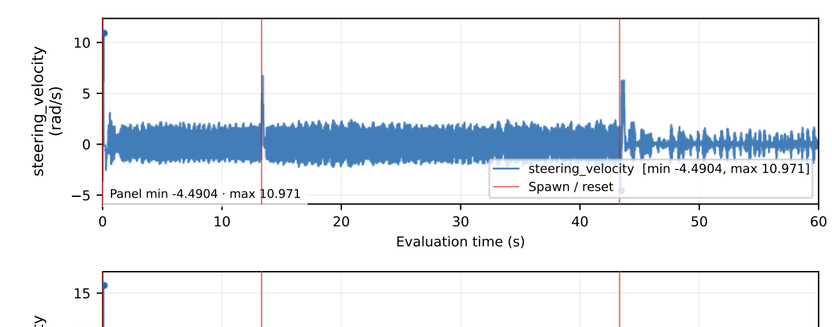



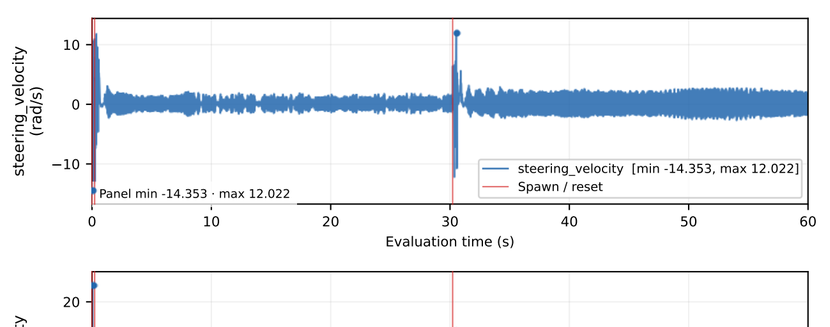



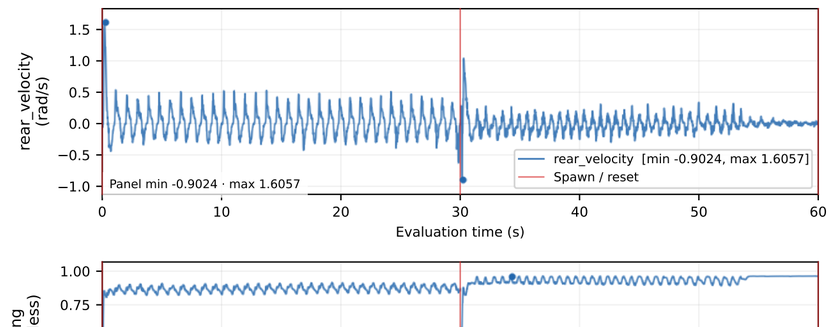



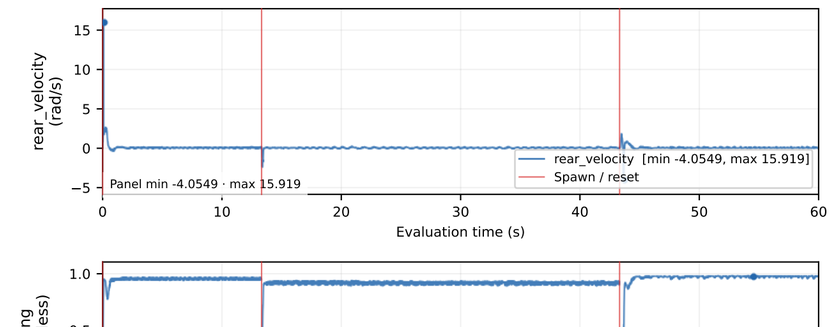



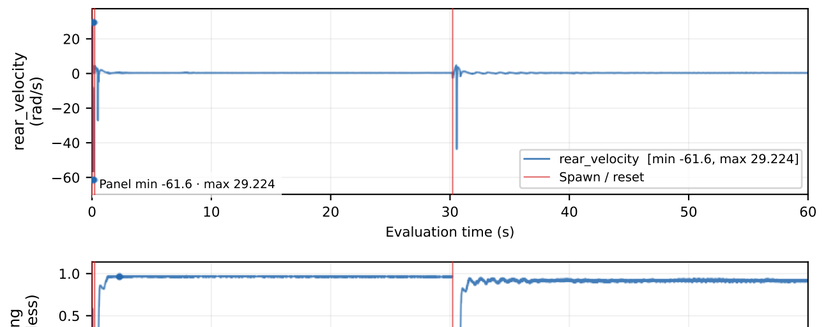



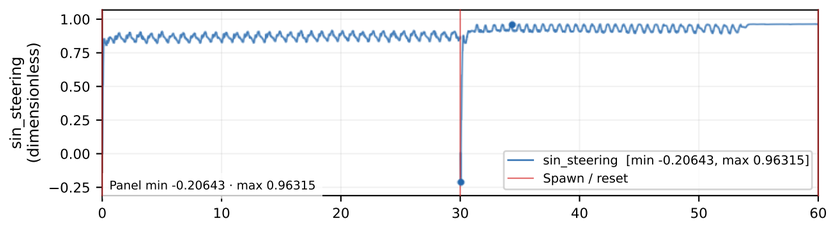



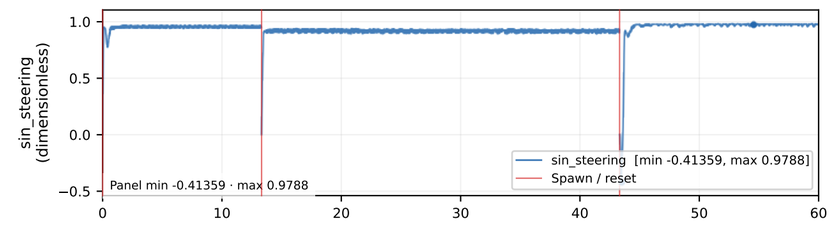



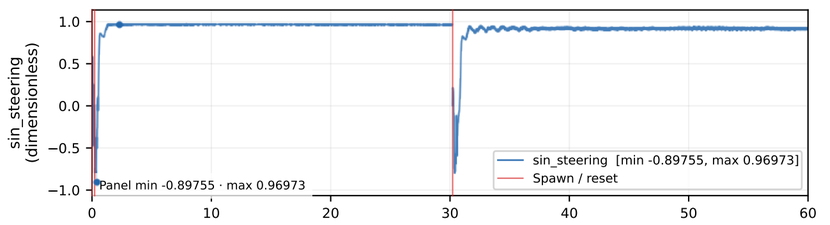



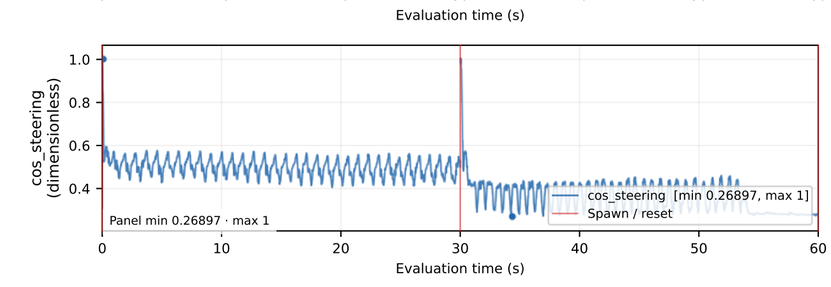



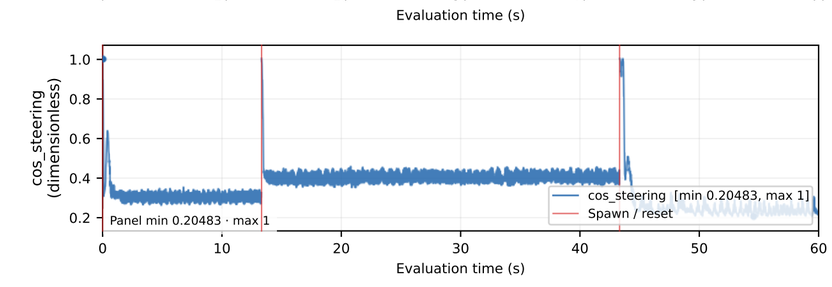



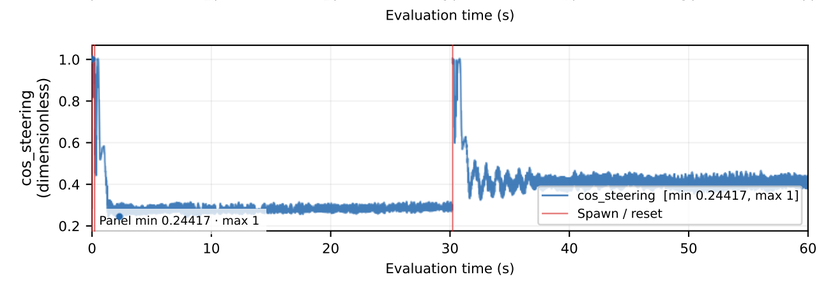




50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

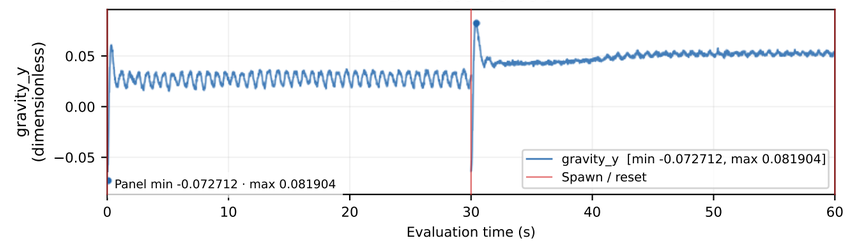
```
SAMPLED DR: env 46; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [0.813902, 0.964713, 1.09081, 0.926487]
  CoM XYZ mm [2.33594, 1.13692, 3.41862]
  mu_s [0.579974, 0.787792, 0.576972, 0.726175]
  mu_d [0.579974, 0.617134, 0.576972, 0.558574]
  Kp [steer,rear] [9.21021, 0]; Kd [0.214087, 0]
  spawn roll 0.0381843 rad; rate -0.107651 rad/s
  gyro bias XYZ rad/s [0.000202893, -0.000185471, -0.000147127]
  gravity bias YZ [0.00189576, 0.0012899]
E2 [30.00, 60.00) s:
  mass x [1.17386, 0.936098, 1.10321, 1.14728]
  CoM XYZ mm [-1.03818, -0.664905, 2.66431]
  mu_s [0.532645, 0.681711, 0.602694, 0.539956]
  mu_d [0.532645, 0.611756, 0.602694, 0.539956]
  Kp [steer,rear] [9.17165, 0]; Kd [0.212853, 0]
  spawn roll 0.00172687 rad; rate -0.246702 rad/s
  gyro bias XYZ rad/s [-5.22129e-05, 0.000184728, 0.00022664]
  gravity bias YZ [0.00270684, -0.00120501]
Six significant digits shown; full precision: sampled_dr_values.json.
```

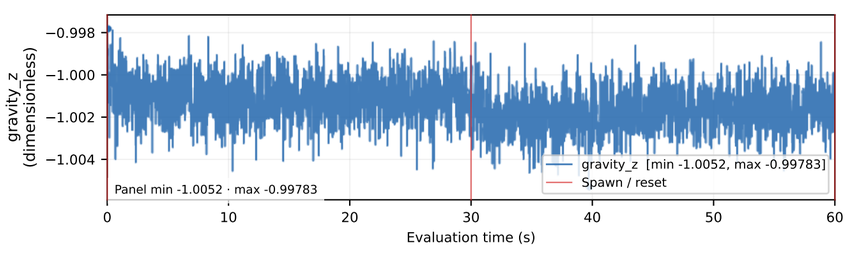
```
SAMPLED DR: env 46; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 13.32) s:
  mass x [0.882498, 1.08128, 1.0754, 1.07555]
  CoM XYZ mm [1.62438, -3.77742, -3.27171]
  mu_s [0.749678, 0.727426, 0.640016, 0.69316]
  mu_d [0.741429, 0.606358, 0.513141, 0.69316]
  Kp [steer,rear] [9.12117, 0]; Kd [0.207069, 0]
  spawn roll 0.0932981 rad; rate -0.0927966 rad/s
  gyro bias XYZ rad/s [0.000126401, 0.000390188, -0.000109424]
  gravity bias YZ [0.00110445, 0.00119378]
E2 [13.32, 43.32) s:
  mass x [1.01361, 0.802286, 0.990541, 1.13248]
  CoM XYZ mm [1.09206, 0.0480632, 1.10722]
  mu_s [0.732187, 0.637675, 0.69316, 0.538434]
  mu_d [0.631664, 0.637675, 0.69316, 0.538434]
  Kp [steer,rear] [10.5326, 0]; Kd [0.195451, 0]
  spawn roll -0.0183043 rad; rate 0.0863785 rad/s
  gyro bias XYZ rad/s [-5.88265e-05, 0.000388015, -0.000227037]
  gravity bias YZ [-0.000566167, 0.00130714]
E3 [43.32, 60.00) s:
  mass x [1.02491, 0.911508, 1.10754, 1.13636]
  CoM XYZ mm [-3.94382, -4.78781, 3.85707]
  mu_s [0.546287, 0.616244, 0.757579, 0.728342]
  mu_d [0.546287, 0.586498, 0.70921, 0.728342]
  Kp [steer,rear] [10.5, 0]; Kd [0.191548, 0]
  spawn roll -0.04388 rad; rate 0.0375422 rad/s
  gyro bias XYZ rad/s [-0.000171093, 0.000145314, 6.84799e-05]
  gravity bias YZ [-0.0004935, 0.00240581]
Six significant digits shown; full precision: sampled_dr_values.json.
```

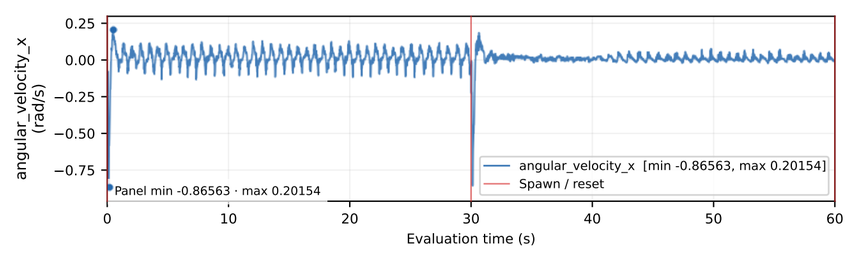
```
SAMPLED DR: env 46; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.22) s:
  mass x [0.882498, 1.08128, 1.0754, 1.07555]
  CoM XYZ mm [1.62438, -3.77742, -3.27171]
  mu_s [0.749678, 0.727426, 0.640016, 0.69316]
  mu_d [0.741429, 0.606358, 0.513141, 0.69316]
  Kp [steer,rear] [9.12117, 0]; Kd [0.207069, 0]
  spawn roll 0.0932981 rad; rate -0.0927966 rad/s
  gyro bias XYZ rad/s [0.000126401, 0.000390188, -0.000109424]
  gravity bias YZ [0.00110445, 0.00119378]
E2 [0.22, 30.22) s:
  mass x [0.885525, 1.18296, 1.07899, 0.9302]
  CoM XYZ mm [-2.16469, -4.08174, -1.2722]
  mu_s [0.732187, 0.640016, 0.69316, 0.640867]
  mu_d [0.631664, 0.513141, 0.69316, 0.556841]
  Kp [steer,rear] [9.38561, 0]; Kd [0.184505, 0]
  spawn roll -0.0790354 rad; rate 0.0140316 rad/s
  gyro bias XYZ rad/s [8.00441e-05, 5.28129e-05, -0.000146935]
  gravity bias YZ [0.00550037, 0.00103566]
E3 [30.22, 60.00) s:
  mass x [1.00945, 0.910747, 0.800438, 0.876145]
  CoM XYZ mm [-2.48635, -0.00111701, -0.126316]
  mu_s [0.728342, 0.723429, 0.568073, 0.538976]
  mu_d [0.728342, 0.723429, 0.568073, 0.538976]
  Kp [steer,rear] [9.47252, 0]; Kd [0.187993, 0]
  spawn roll -0.046428 rad; rate 0.187411 rad/s
  gyro bias XYZ rad/s [0.000288874, -0.000303692, 9.03436e-05]
  gravity bias YZ [0.000921866, 0.00245169]
Six significant digits shown; full precision: sampled_dr_values.json.
```

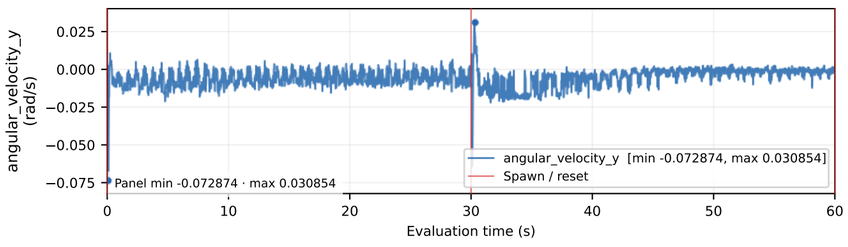
# model_2500 · All 11 actual policy observations · env 47 · full 60 s

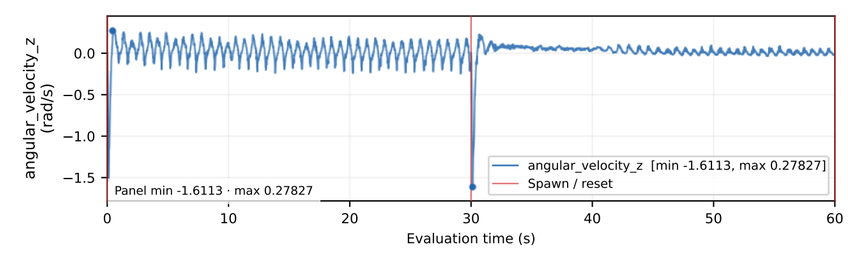
Isaac Lab · physics/PD 400 Hz · env 47
DR · 0.00–60.00 s

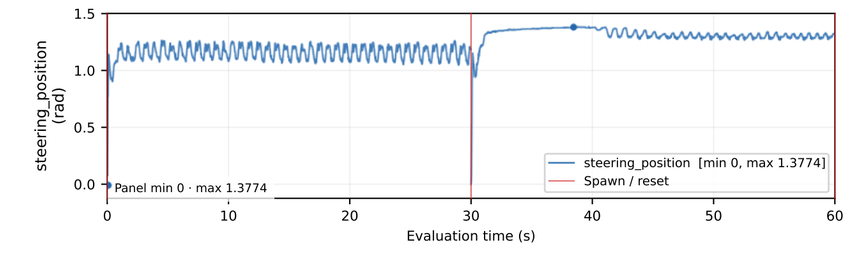


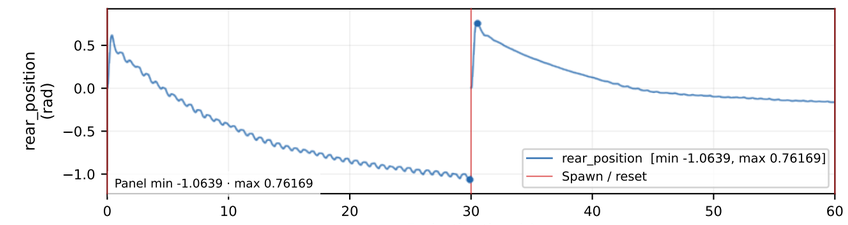



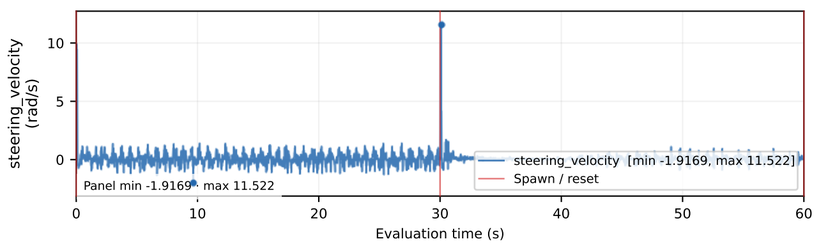



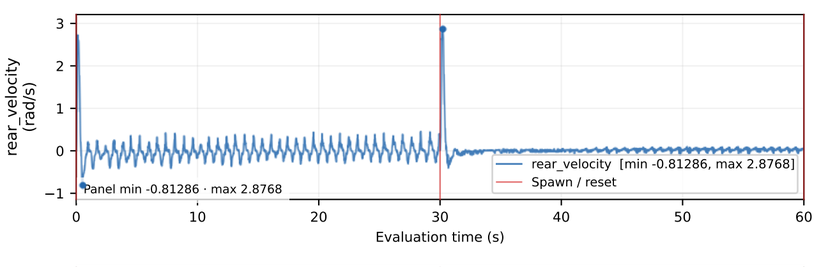



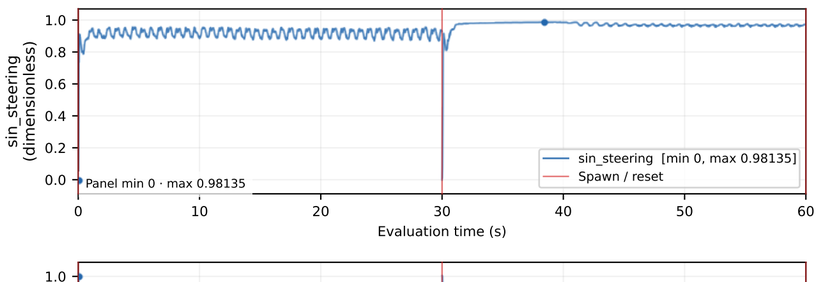



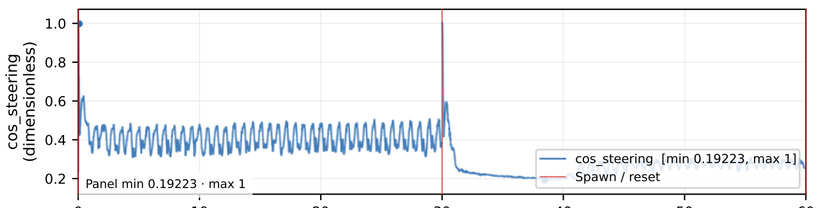



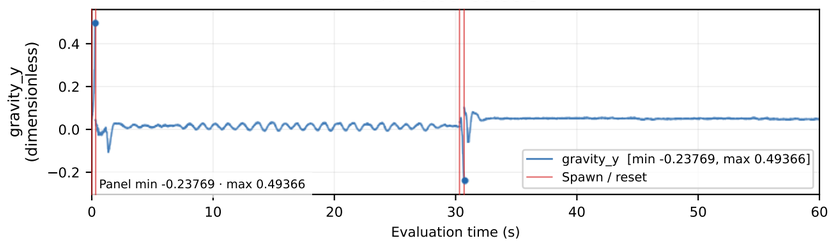



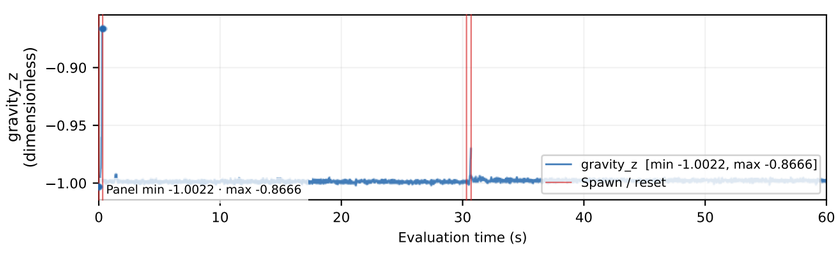



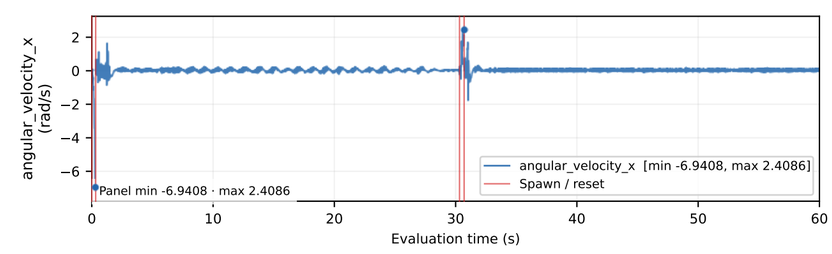



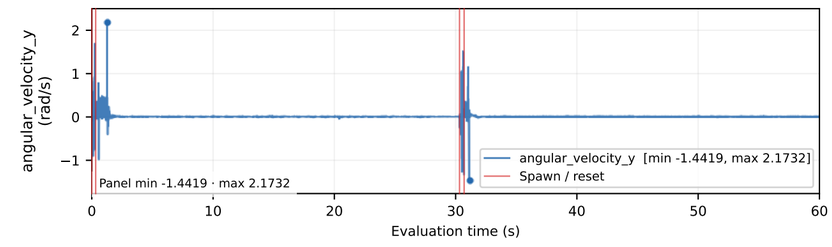



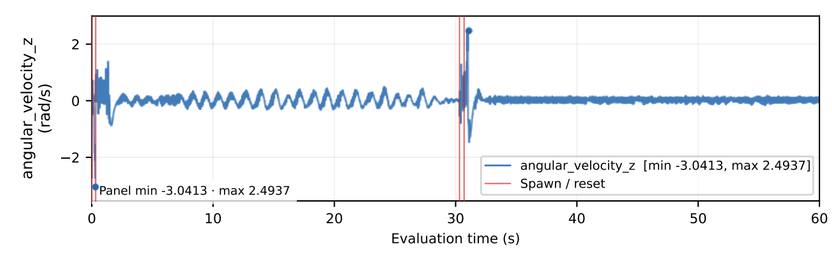

MuJoCo · physics/PD 400 Hz · env 47
DR · 0.00–60.00 s

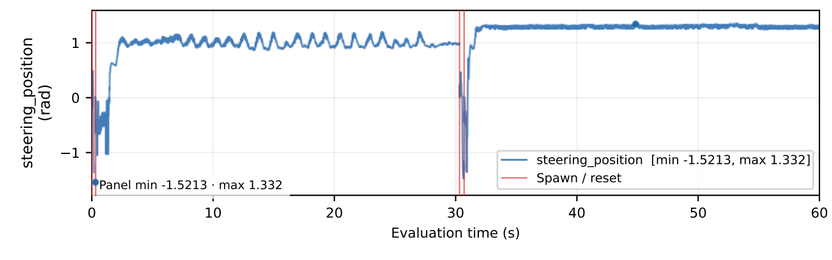


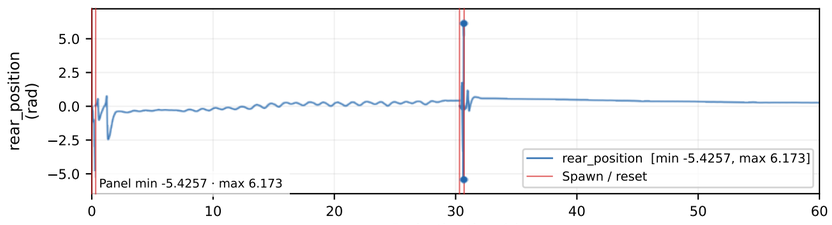



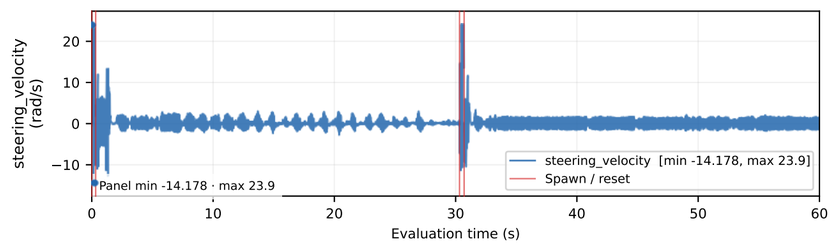



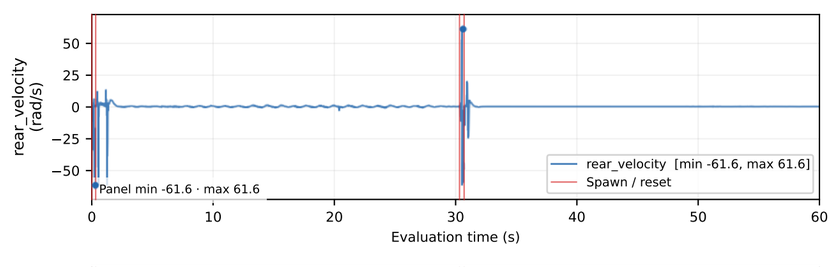



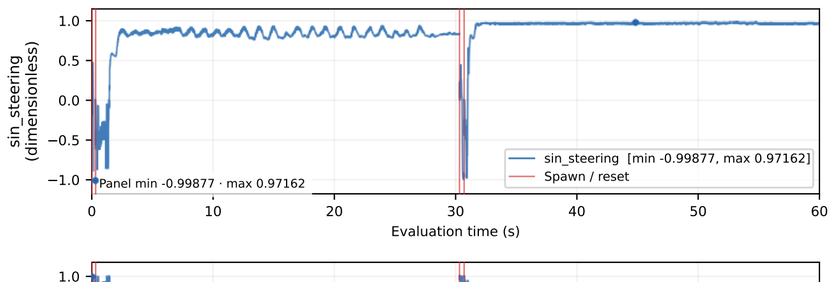



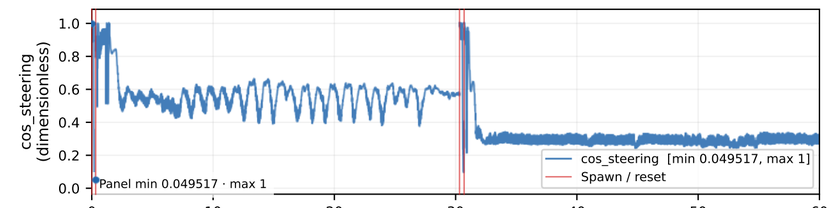



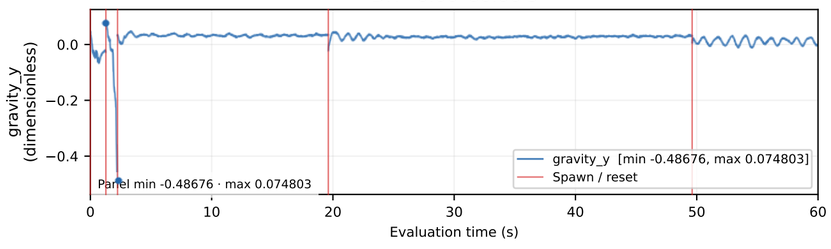



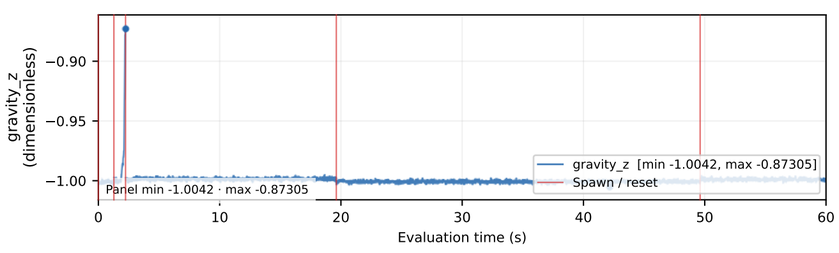



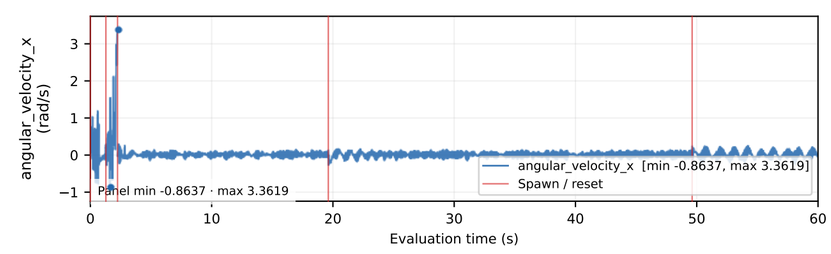



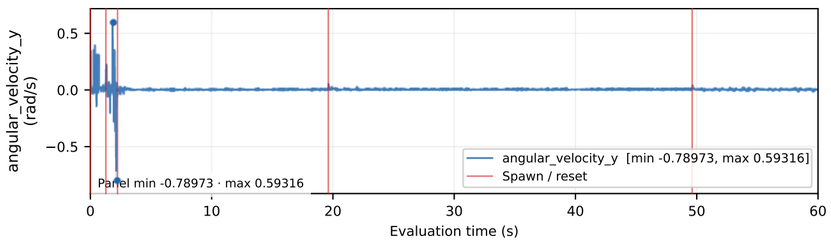



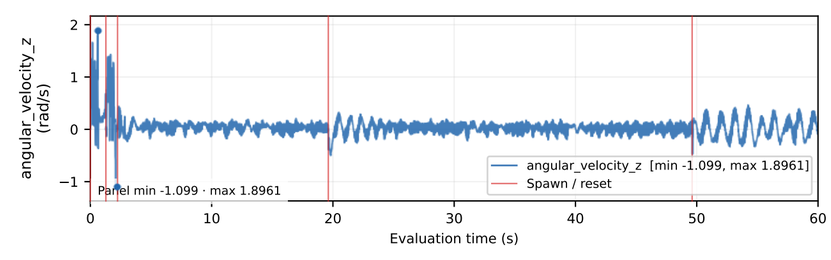

MuJoCo · physics/PD 800 Hz · env 47
DR · 0.00–60.00 s

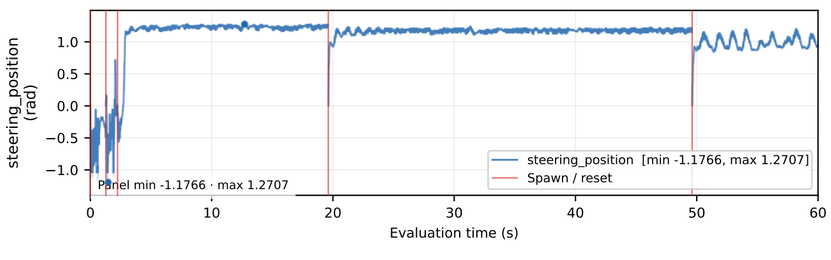


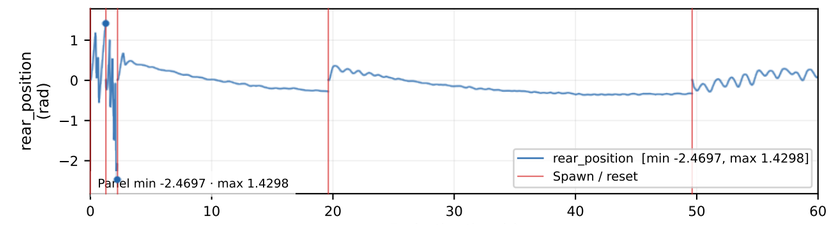



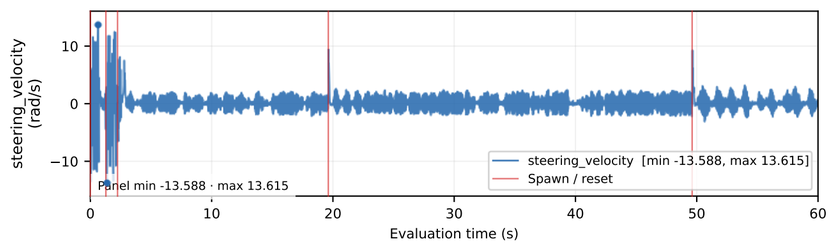



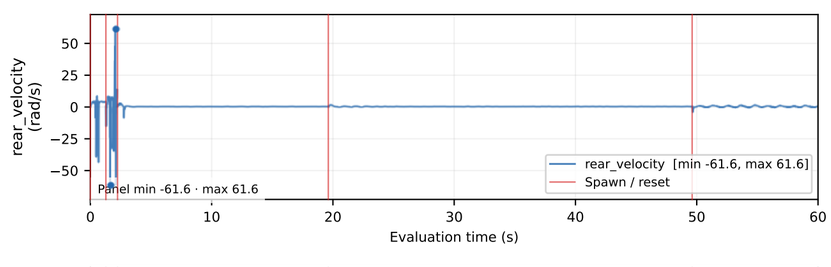



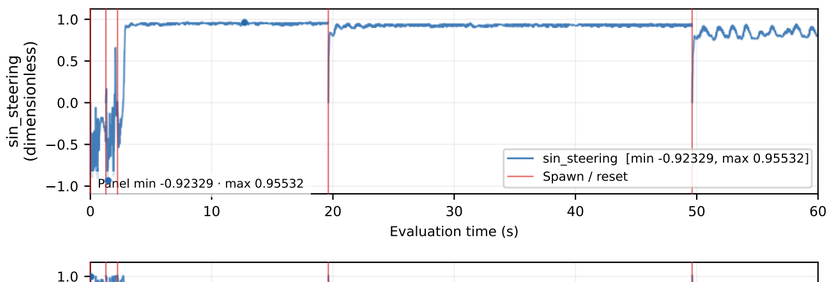



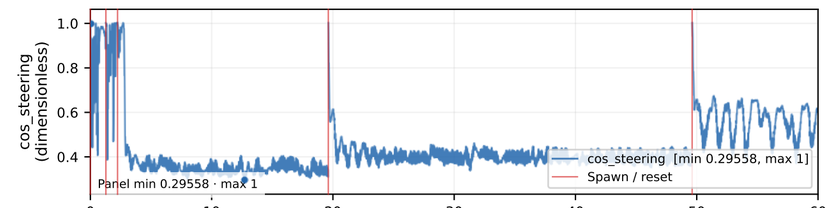




50 Hz policy · seed 42 · red = every spawn/reset in view · dots + legends = recorded minima/maxima
Independent full-range Y axes · no smoothing, numerical clipping or response time shifting · DR draws are not paired across engines.

```
SAMPLED DR: env 47; body order: base, rear_wheel_1, steering_1, front_wheel_1
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
Recovered with all 18 original capture arrays byte-identical.
E1 [0.00, 30.00) s:
  mass x [1.03655, 1.17511, 1.11408, 0.877524]
  CoM XYZ mm [-4.44587, -0.567719, -2.63108]
  mu_s [0.760821, 0.726175, 0.560535, 0.726175]
  mu_d [0.670315, 0.558574, 0.560535, 0.558574]
  Kp [steer,rear] [10.6797, 0]; Kd [0.214648, 0]
  spawn roll 0.0742524 rad; rate -0.213024 rad/s
  gyro bias XYZ rad/s [-4.5567e-05, 0.000271351, -5.81968e-05]
  gravity bias YZ [0.00115448, -0.00144247]
E2 [30.00, 60.00) s:
  mass x [0.899864, 0.877441, 1.16899, 1.08508]
  CoM XYZ mm [-2.47404, -3.20068, -4.93082]
  mu_s [0.785466, 0.647391, 0.713129, 0.693205]
  mu_d [0.52258, 0.647391, 0.713129, 0.693205]
  Kp [steer,rear] [9.54417, 0]; Kd [0.213168, 0]
  spawn roll 0.0653071 rad; rate -0.227627 rad/s
  gyro bias XYZ rad/s [-5.883e-05, 0.000143332, 0.000143997]
  gravity bias YZ [0.000960929, -0.00292982]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 47; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 0.32) s:
  mass x [1.03031, 1.10798, 1.09245, 0.807864]
  CoM XYZ mm [0.400709, 4.01135, 2.78235]
  mu_s [0.528253, 0.71008, 0.736077, 0.748289]
  mu_d [0.528253, 0.59371, 0.699455, 0.689499]
  Kp [steer,rear] [9.35461, 0]; Kd [0.21788, 0]
  spawn roll -0.0468968 rad; rate -0.0970046 rad/s
  gyro bias XYZ rad/s [3.01175e-06, -0.000120107, 6.87117e-05]
  gravity bias YZ [-0.00422143, -0.00194905]
E2 [0.32, 30.32) s:
  mass x [0.819334, 0.975526, 1.00746, 0.983347]
  CoM XYZ mm [-2.60108, 1.40927, 2.02062]
  mu_s [0.597748, 0.700521, 0.757579, 0.71008]
  mu_d [0.597748, 0.641329, 0.70921, 0.59371]
  Kp [steer,rear] [9.07265, 0]; Kd [0.185688, 0]
  spawn roll -0.0440747 rad; rate 0.0112541 rad/s
  gyro bias XYZ rad/s [3.78981e-05, 2.99002e-05, 8.45185e-05]
  gravity bias YZ [-0.00151106, 0.00107793]
E3 [30.32, 30.70) s:
  mass x [0.935038, 1.02052, 1.12262, 0.909411]
  CoM XYZ mm [1.74958, -3.78616, 4.51876]
  mu_s [0.704749, 0.69316, 0.597748, 0.538434]
  mu_d [0.541926, 0.69316, 0.597748, 0.538434]
  Kp [steer,rear] [10.458, 0]; Kd [0.211661, 0]
  spawn roll -0.0330262 rad; rate 0.160557 rad/s
  gyro bias XYZ rad/s [-0.000294888, -9.13743e-05, 0.000444933]
  gravity bias YZ [-6.74809e-05, 0.00103594]
E4 [30.70, 60.00) s:
  mass x [0.901264, 1.02977, 1.04103, 1.05278]
  CoM XYZ mm [-0.580672, -3.41664, 3.29062]
  mu_s [0.568073, 0.666375, 0.704749, 0.736077]
  mu_d [0.568073, 0.519145, 0.541926, 0.699455]
  Kp [steer,rear] [10.2127, 0]; Kd [0.186575, 0]
  spawn roll -0.0988457 rad; rate 0.195537 rad/s
  gyro bias XYZ rad/s [4.37454e-05, 0.000172272, 8.1821e-05]
  gravity bias YZ [0.00021133, 0.00109951]
Six significant digits shown; full precision: sampled_dr_values.json.
```

```
SAMPLED DR: env 47; body order: base, rear_wheel, steering, front_wheel
Contact arrays: collision-shape order; restitution = 0.
CoM XYZ: shared Isaac-link-frame offset (mm); gravity bias: unit gravity.
mu_d is sampled metadata; MuJoCo's native contact cone uses mu_s.
E1 [0.00, 1.28) s:
  mass x [1.03031, 1.10798, 1.09245, 0.807864]
  CoM XYZ mm [0.400709, 4.01135, 2.78235]
  mu_s [0.528253, 0.71008, 0.736077, 0.748289]
  mu_d [0.528253, 0.59371, 0.699455, 0.689499]
  Kp [steer,rear] [9.35461, 0]; Kd [0.21788, 0]
  spawn roll -0.0468968 rad; rate -0.0970046 rad/s
  gyro bias XYZ rad/s [3.01175e-06, -0.000120107, 6.87117e-05]
  gravity bias YZ [-0.00422143, -0.00194905]
E2 [1.28, 2.24) s:
  mass x [0.988684, 0.904768, 0.895446, 1.12629]
  CoM XYZ mm [-0.0644951, 2.06511, -3.90885]
  mu_s [0.71008, 0.71008, 0.69316, 0.538976]
  mu_d [0.59371, 0.59371, 0.69316, 0.538976]
  Kp [steer,rear] [10.5017, 0]; Kd [0.205067, 0]
  spawn roll -0.0737216 rad; rate 0.189624 rad/s
  gyro bias XYZ rad/s [-3.92332e-06, 0.000134535, -0.000308125]
  gravity bias YZ [4.32135e-05, -0.00068459]
E3 [2.24, 19.62) s:
  mass x [1.08582, 1.16874, 0.998601, 1.03475]
  CoM XYZ mm [-4.1558, -1.24468, 4.9313]
  mu_s [0.748289, 0.611239, 0.528253, 0.597748]
  mu_d [0.689499, 0.611239, 0.528253, 0.597748]
  Kp [steer,rear] [9.50513, 0]; Kd [0.195058, 0]
  spawn roll -0.0301954 rad; rate -0.0384992 rad/s
  gyro bias XYZ rad/s [2.39944e-06, -9.88794e-05, 0.000250088]
  gravity bias YZ [0.00208184, 0.000981116]
E4 [19.62, 49.62) s:
  mass x [1.1446, 1.15252, 1.16758, 1.04264]
  CoM XYZ mm [-4.53675, -0.501051, 2.79191]
  mu_s [0.71008, 0.757579, 0.538976, 0.728342]
  mu_d [0.59371, 0.70921, 0.538976, 0.728342]
  Kp [steer,rear] [9.44164, 0]; Kd [0.18992, 0]
  spawn roll 0.0247201 rad; rate 0.114003 rad/s
  gyro bias XYZ rad/s [-0.000271496, 0.000279174, 0.000242257]
  gravity bias YZ [0.0008539, -0.00111884]
E5 [49.62, 60.00) s:
  mass x [0.909109, 1.07095, 1.03261, 1.03408]
  CoM XYZ mm [-3.01552, 1.72354, -4.71103]
  mu_s [0.748289, 0.71155, 0.71008, 0.637675]
  mu_d [0.689499, 0.71155, 0.59371, 0.637675]
  Kp [steer,rear] [9.56325, 0]; Kd [0.208263, 0]
  spawn roll -0.00541428 rad; rate 0.220684 rad/s
  gyro bias XYZ rad/s [0.000146279, -5.36009e-05, 3.81239e-05]
  gravity bias YZ [-0.000445671, 0.000948283]
Six significant digits shown; full precision: sampled_dr_values.json.
```